

POWERFACTORY

PowerFactory 2024

Technical Reference

Common Result Variables for Terminals and Elements

Harmonics Analysis

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

The Harmonic Load Flow calculation in *PowerFactory* provides users with a multitude of result (or 'monitor') variables to choose from. The calculation of these variables in PF is described in this document, for three phase systems, considering harmonic sources defined as 'IEC' (see [3]) and 'non-IEC'.

2 Calculation Considering Only Non-IEC Harmonic Sources

2.1 Balanced Calculation

2.1.0.1 Branch Monitor Variables

Table 2.1 provides the basic nomenclature used throughout this section.

Table 2.1: Nomenclature

Name	Unit	Description
f_k	Hz	Frequency of harmonic order k
f_{out}	Hz	Output frequency specified by user in Harmonic Load Flow Command (parameter name: *ifshow*)
$\underline{i}(f_k)$	p.u.	Complex harmonic current at frequency of harmonic order k (all p.u. values are based on I_{nom})
$\underline{u}(f_k)$	p.u.	Complex harmonic voltage at frequency of harmonic order k (all p.u. values are based on U_{nom})
$\underline{I}(f_k)$	A	Complex harmonic current, at frequency of harmonic order k
$\underline{U}(f_k)$	kV	Complex harmonic voltage, at frequency of harmonic order k
I_{nom}, i_{nom}	A, p.u.	Nominal current of element
U_{nom}, u_{nom}	kV, p.u.	Rated voltage of element
i_{max}	p.u.	Maximum current of element
$tif(f_k)$	-	Telephone influence factor (according to IEEE) at frequency of harmonic order k

Monitor Variables for Output Frequency The following monitor variables are calculated for user-selected *Output Frequency*, f_{out}:

Intermediate variables:

$\underline{S} = \sqrt{3} \cdot \underline{U} \cdot \underline{I}^*$	Complex power in kVA

Monitor variables:

$i = \lvert\underline{i}\rvert$	Current magnitude in p.u.
$I = i \cdot I_{nom}$	Current magnitude in A
$I1 = I$	Positive sequence current magnitude in A
$I1 = 0$	Positive sequence current magnitude in A, for harmonic orders 5, 11, 17, 23,...
$I2 = I$	Negative sequence current magnitude in A, for harmonic orders 5, 11, 17, 23,...
$I2 = 0$	Negative sequence current magnitude in A
$phiui = \phi_{\underline{S}}$	Angle between voltage and current in deg
$phii = \phi_{\underline{i}}$	Current angle, absolute in deg
$P = \Re(\underline{S})$	Active power in kW
$Q = \Im(\underline{S})$	Reactive power in kvar
$S = \lvert\underline{S}\rvert$	Apparent power in kVA
$cosphi = \frac{P}{S}$	Power factor

Calculation variables: The following calculation variables are available for lines, transformers and series compensation:

$Losses = Re(\underline{S})$	Losses in kW
$loading = i_{max}/i_{nom} \cdot 100$	Loading in %

The following calculation variable is available for machines (static generator, asynchronous machine and synchronous machine):

$loading = S/S_n \cdot 100$	Loading in %, where S_n is the total apparent nominal power in MVA.

Monitor Variables Considering All Frequencies: The following variables are calculated considering all frequencies, f_k:

Intermediate variables:

$i(f_k) = \|\underline{i}(f_k)\|$	Current magnitude in p.u.
$\underline{S}(f_k) = \sqrt{3} \cdot \underline{U}(f_k) \cdot \underline{I}(f_k)^*$	Complex power in kVA
$P(f_k) = Re(\underline{S}(f_k))$	Active power in kW

Monitor variables:

$HD = \frac{i(f_{out})}{i(f_1)} \cdot 100$	Harmonic distortion, current in %; based on fundamental frequency values
$HD = \frac{i(f_{out})}{i_{nom}} \cdot 100$	Harmonic distortion, current in %; based on nominal current
$HF = \frac{i(f_{out})}{irms} \cdot 100$	Harmonic factor, current in %
$Irms = \sqrt{\sum_{k=1}^{n} i(f_k)^2} \cdot I_{nom}$	Current, RMS in kA
$irms = \frac{Irms}{I_{nom}}$	Current, RMS in p.u.
$THD = \frac{1}{i(f_1)} \cdot \sqrt{irms^2 - i(f_1)^2} \cdot 100$	Total harmonic distortion, current in %; based on fundamental frequency values
$THD = \frac{1}{i_{nom}} \cdot \sqrt{irms^2 - i(f_1)^2} \cdot 100$	Total harmonic distortion, current in %; based on nominal current
$THF = \frac{1}{irms} \cdot \sqrt{irms^2 - i(f_1)^2} \cdot 100$	Total harmonic factor, current in %
$TP = \sum_{k=1}^{n} P(f_k)$	Total active power in MW
$TS = \sqrt{Irms^2 \cdot Urms^2 \cdot 3}$	Total apparent power in MVA, where $Urms = \sqrt{\sum_{k=1}^{n} \|\underline{u}(f_k)\|^2} \cdot U_{nom}$
$TQ = \sqrt{TS^2 - TP^2}$	Total reactive power in Mvar
$Tcosphi = \frac{TP}{TS}$	Total power factor
$IT = \sqrt{\sum_{k=1}^{n} tif(f_k)^2 \cdot i(f_k)^2} \cdot I_{nom}$	IT-product in kA
$TAD = \frac{1}{i(f_1)} \cdot \sum_{k=1}^{n} [i(f_k) - i(f_1)] \cdot 100$	Total arithmetic distortion in %

Calculation variables: The following calculation variables are available for lines, transformers, series compensation:

$LossesTot = \sum_{h\geq 1}^{H} Losses$	Total losses, including the fundamental in kW, where h is the harmonic order
$LossesHrm = \sum_{h>1}^{H} Losses$	Harmonic losses in kW
$loadingTot = irms_{max}/i_{nom} \cdot 100$	Total loading in %

2.1.0.2 Bus Monitor Variables

Table 2.2 provides the basic nomenclature used throughout this section.

Table 2.2: Nomenclature

Name	Unit	Description
f_k	Hz	Frequency of harmonic order k
f_{out}	Hz	Output frequency specified by user in Harmonic Load Flow Command (parameter name: *ifshow*)
$\underline{u}(f_k)$		Complex harmonic voltage at frequency of harmonic order k
$u(f_k)$	p.u.	Voltage magnitude at frequency of harmonic order k
U_{nom}	kV	Nominal voltage of the busbar
$\phi_{initial}$	deg	Initial voltage angle
$\underline{Z}$	Ω	Short-circuit impedance (positive or negative sequence, depending on the sequence characteristic of the harmonic order). In the balanced Frequency Sweep, it is the positive sequence impedance.
$\underline{Z}_{nc}$	Ω	Short-circuit impedance without capacitive effects (only available for the Frequency Sweep calculation)

Monitor Variables for Output Frequency The following monitor variables are calculated for user-selected output frequency, f_{out}:

$R = Re\,(\underline{Z})$	Network resistance in Ω
$X = Im\,(\underline{Z})$	Network reactance in Ω
$Z = \lvert\underline{Z}\rvert$	Network impedance magnitude in Ω
$phiz = \phi_{\underline{Z}}$	Network impedance angle in deg
$u = \lvert\underline{u}\rvert$	Voltage magnitude in p.u.
$upc = u \cdot 100$	Voltage magnitude in %

$u1 = u$	Voltage magnitude in p.u.
$u1 = 0$	Voltage magnitude in p.u., for orders 5, 11, 17, 23,. . . ,
$u1pc = u1 \cdot 100$	Voltage magnitude in %
$U = u \cdot U_{nom}/\sqrt{3}$	Line-ground voltage magnitude in kV
$Ul = u \cdot U_{nom}$	Line-line voltage magnitude in kV
$phiu = \phi_{\underline{u}}$	Voltage angle in deg
$phiurel = \phi_{\underline{u}} - \phi_{initial}$	Voltage, relative angle in deg
$du = upc - 100$	Voltage deviation in %
$u2 = u$	Negative sequence voltage in p.u., for orders 5, 11, 17, 23, . . . ,
$u2 = 0$	Negative sequence voltage in p.u.

and the following quantities for the short-circuit impedance without capacitive effects (Z_{nc}). The quantities are only available for the Frequency Sweep calculation when one of the following calculation variables is selected for recording, otherwise the value is set to zero.

$Rnc = Re\,(\underline{Z}_{nc})$	Network resistance (without C) in Ω
$Xnc = Im\,(\underline{Z}_{nc})$	Network reactance (without C) in Ω
$Znc = \lvert\underline{Z}_{nc}\rvert$	Network impedance magnitude (without C) in Ω
$phiznc = \phi_{\underline{Z}_{nc}}$	Network impedance angle (without C) in deg

Monitor Variables Considering All Frequencies: The following variables are calculated considering all frequencies, f_k:

Intermediate variables:

$u(f_k) = \lvert\underline{u}(f_k)\rvert$	Voltage magnitude in p.u.

Monitor variables:

$urms = \sqrt{\sum_{k=1}^{n} u(f_k)^2}$	RMS value of voltage in p.u.
$HD = \frac{u(f_{out})}{u(f_1)} \cdot 100$	Harmonic distortion in %; based on fundamental frequency values
$HD = \frac{u(f_{out})}{u_{nom}} \cdot 100$	Harmonic distortion in %; Based on nominal voltage, where $u_{nom} = 1$p.u.

$HF = \frac{u(f_{out})}{urms} \cdot 100$	Harmonic factor in %
$THD = \frac{1}{u(f_1)} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic distortion in %; based on fundamental frequency values
$THD = \frac{1}{u_{nom}} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic distortion in %; based on nominal voltage, where $u_{nom} = 1$ p.u.
$THF = \frac{1}{urms} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic factor in %
$Ulrms = urms \cdot U_{nom}$	RMS value of line-line voltage in kV
$Urms = urms \cdot U_{nom}/\sqrt{3}$	RMS value of line-ground voltage in k)
$TAD = \frac{1}{u(f_1)} \cdot [\sum_{k=1}^{n} u(f_k) - u(f_1)] \cdot 100$	Total arithmetic distortion in %
$uasum = \sum_{k=1}^{n} \lvert \underline{u}(f_k) \rvert$	Arithmetic voltage sum in p.u.
$TIF = \frac{1}{urms} \sqrt{\sum_{k=1}^{n} tif(f_k)^2 \cdot u(f_k)^2}$	Telephone interference factor
$urmsint = \sqrt{\sum_{k=2}^{n} u(f_k)^2}$	RMS value of integer harmonics in p.u.
$uasumint = \sum_{k=2}^{n} u(f_k)$	Arithmetic sum of integer harmonics in p.u.
$THDint = THD$	Total harmonic distortion in %, integer harmonic orders
$THDnint = 0$	Total harmonic distortion in %, non-integer harmonic orders
$TADint = TAD$	Arithmetic distortion in %, integer harmonic orders
$TADnint = 0$	Arithmetic distortion in %, non-integer harmonic orders

2.2 Unbalanced Calculation

2.2.0.1 Branch Monitor Variables

Table 2.3 provides the basic nomenclature used throughout this section.

Table 2.3: Nomenclature

Name	Unit	Description
f_k	Hz	Frequency at harmonic order k
f_{out}	Hz	Output frequency specified by user in Harmonic Load Flow Command (parameter name: *ifshow*)
$\underline{i}_a(f_k)$	p.u.	Complex harmonic current for phase A, at frequency of harmonic order k (all p.u. values are based on I_{nom}). Similar nomenclature follows for other phases.
$\underline{u}_a(f_k)$	p.u.	Complex harmonic voltage in p.u. for phase B, at frequency of harmonic order k (all p.u. values are based on U_{nom}). Similar nomenclature follows for other phases.
$\underline{I}_a(f_k)$	A	Complex harmonic current for phase A, at frequency of harmonic order k. Similar nomenclature follows for other phases.
$\underline{U}_a(f_k)$	kV	Complex harmonic voltage for phase A, at frequency of harmonic order k. Similar nomenclature follows for other phases.
$\underline{i}_0(f_k)$	p.u.	Complex zero sequence current, at frequency of harmonic order k.
$\underline{u}_0(f_k)$	p.u.	Complex zero sequence voltage, at frequency of harmonic order k.
$i{:}A(f_k)$	p.u.	Current magnitude at frequency of harmonic order k. Similar nomenclature follows for other phases.
$i0(f_k)$	p.u.	Zero sequence current magnitude at frequency of harmonic order k. Similar nomenclature follows for positive and negative sequence currents.
I_{nom}, i_{nom}	A; p.u.	Nominal current of element
U_{nom}, u_{nom}	kV; p.u.	Rated voltage of element
$tif(f_k)$		Telephone influence factor (according to IEEE) at frequency of harmonic order k.

Monitor Variables for Output Frequency The following monitor variables are calculated for user-selected output frequency, f_{out}:

Intermediate variables:

$\underline{S}_a = \underline{U}_a \cdot \underline{I}_a^*$	Complex power in kVA
$\underline{S}_b = \underline{U}_b \cdot \underline{I}_b^*$	
$\underline{S}_c = \underline{U}_c \cdot \underline{I}_c^*$	
$\underline{S}_n = \underline{U}_n \cdot \underline{I}_n^*$	

Monitor variables:

$i{:}A = \lvert\underline{i}_a\rvert$	Phase currents in p.u. based on rated current of element
$i{:}B = \lvert\underline{i}_b\rvert$	
$i{:}C = \lvert\underline{i}_c\rvert$	
$i{:}N = \lvert\underline{i}_n\rvert$	
$I{:}A = i{:}A \cdot I_{nom}$	Phase currents in A
$I{:}B = i{:}B \cdot I_{nom}$	
$I{:}C = i{:}C \cdot I_{nom}$	
$I{:}N = i{:}N \cdot I_{nom}$	
$phiui{:}A = \phi_{\underline{S}_a}$	Angle between voltage and current in deg
$phiui{:}B = \phi_{\underline{S}_b}$	
$phiui{:}C = \phi_{\underline{S}_c}$	
$phiui{:}N = \phi_{\underline{S}_n}$	
$phii{:}A = \phi_{\underline{i}_a}$	Current angle, absolute in deg
$phii{:}B = \phi_{\underline{i}_b}$	
$phii{:}C = \phi_{\underline{i}_c}$	
$phii{:}N = \phi_{\underline{i}_n}$	

$P{:}A = Re\left(\underline{S}_a\right)$	Active power in kW
$P{:}B = Re\left(\underline{S}_b\right)$	
$P{:}C = Re\left(\underline{S}_c\right)$	
$P{:}N = Re\left(\underline{S}_n\right)$	
$Q{:}A = Im\left(\underline{S}_a\right)$	Reactive power in kvar
$Q{:}B = Im\left(\underline{S}_b\right)$	
$Q{:}C = Im\left(\underline{S}_c\right)$	
$Q{:}N = Im\left(\underline{S}_n\right)$	
$S{:}A = \lvert\underline{S}_a\rvert$	Apparent power in kVA
$S{:}B = \lvert\underline{S}_b\rvert$	
$S{:}C = \lvert\underline{S}_c\rvert$	
$S{:}N = \lvert\underline{S}_n\rvert$	
$cosphi{:}A = \frac{P:A}{S:A}$	Power factor
$cosphi{:}B = \frac{P:B}{S:B}$	
$cosphi{:}C = \frac{P:C}{S:C}$	
$cosphi{:}N = \frac{P:N}{S:N}$	
$i0 = \frac{1}{3}\left(i{:}A + i{:}B + i{:}C\right)$	Zero sequence current in p.u.
$I0 = i0 \cdot I_{nom}$	Zero sequence current in A
$phiu0i0 = \phi_{\underline{u}_0} - \phi_{\underline{i}_0}$	Angle between voltage and current in zero sequence system in deg
$i1 = \frac{1}{3}\left(i{:}A + a \cdot i{:}B + a^2 \cdot i{:}C\right)$	Positive sequence current in p.u., where $a = \angle 120°$
$I1 = i1 \cdot I_{nom}$	Positive sequence current in A
$phiu1i1 = \phi_{\underline{u}_1} - \phi_{\underline{i}_1}$	Angle between voltage and current in positive sequence system in deg

$i2 = \frac{1}{3}\left(i{:}A + a^2 \cdot i{:}B + a \cdot i{:}C\right)$	Negative sequence current in p.u.
$I2 = i2 \cdot I_{nom}$	Negative sequence current in A
$phiu2i2 = \phi_{\underline{u}_2} - \phi_{\underline{i}_2}$	Angle between voltage and current in negative sequence system in deg
$Psum = Re\left[\sum_{x=[a,b,c,n]} \underline{S}_x\right]$	Active power in kW
$Qsum = Im\left[\sum_{x=[a,b,c,n]} \underline{S}_x\right]$	Reactive power in kvar
$Ssum = \left\lvert\sum_{x=[a,b,c,n]} \underline{S}_x\right\rvert$	Apparent power in kVA
$cosphisum = \frac{Psum}{Ssum}$	Power factor
$ubfac = \frac{i2}{i1}$	Unbalance factor

Calculation variables: The following calculation variables are available for lines:

$LossesPh{:}A = Re(\underline{S}_a)$	Losses in kW
$LossesPh{:}B = Re(\underline{S}_b)$	
$LossesPh{:}C = Re(\underline{S}_c)$	
$LossesPh{:}N = Re(\underline{S}_n)$	
$Losses = \sum_{x=[A,B,C,N]} LossesPh{:}x$	Losses in kW
$loadingPh{:}A = i{:}A_{max}/i_{nom} \cdot 100$	Loading in %
$loadingPh{:}B = i{:}B_{max}/i_{nom} \cdot 100$	
$loadingPh{:}C = i{:}C_{max}/i_{nom} \cdot 100$	
$loadingPh{:}N = i{:}N_{max}/i_{nom} \cdot 100$	
$loading = i_{max}/i_{nom} \cdot 100$	Loading in %

The following calculation variables are available for transformers and series compensation:

$Losses = Re(\underline{S})$	Losses in kW
$loading = i_{max}/i_{nom} \cdot 100$	Loading in %

The following calculation variable is available for machines (static generator, asynchronous machine and synchronous machine):

$loading = S/S_n \cdot 100$	Loading in %, where S_n is the total apparent nominal power in MVA.

Monitor Variables Considering All Frequencies: The following monitor variables are calculated considering all frequencies f_k:

Intermediate variables:

$i{:}A(f_k) = \lvert \underline{i}_a(f_k) \rvert$	Current magnitude in p.u.
$i{:}B(f_k) = \lvert \underline{i}_b(f_k) \rvert$	
$i{:}C(f_k) = \lvert \underline{i}_c(f_k) \rvert$	
$i{:}N(f_k) = \lvert \underline{i}_n(f_k) \rvert$	
$\underline{S}_a(f_k) = \underline{U}_a(f_k) \cdot \underline{I}_a(f_k)^*$	Complex power in kVA
$\underline{S}_b(f_k) = \underline{U}_b(f_k) \cdot \underline{I}_b(f_k)^*$	
$\underline{S}_c(f_k) = \underline{U}_c(f_k) \cdot \underline{I}_c(f_k)^*$	
$\underline{S}_n(f_k) = \underline{U}_n(f_k) \cdot \underline{I}_n(f_k)^*$	
$P{:}A(f_k) = Re(\underline{S}_a(f_k))$	Active power in kW
$P{:}B(f_k) = Re(\underline{S}_b(f_k))$	
$P{:}C(f_k) = Re(\underline{S}_c(f_k))$	
$P{:}N(f_k) = Re(\underline{S}_n(f_k))$	
$\underline{i0}(f_k) = \frac{1}{3}\left(\underline{i}_a(f_k) + \underline{i}_b(f_k) + \underline{i}_c(f_k)\right)$	Zero sequence current in p.u.
$\underline{i1}(f_k) = \frac{1}{3}\left(i{:}A(f_k) + a \cdot i{:}B(f_k) + a^2 \cdot i{:}C(f_k)\right)$	Positive sequence current in p.u., where $a = \angle 120°$
$\underline{i2}(f_k) = \frac{1}{3}\left(\underline{i}_a(f_k) + a^2 \cdot \underline{i}_b(f_k) + a \cdot \underline{i}_c(f_k)\right)$	Negative sequence current in p.u.
$i0(f_k) = \lvert \underline{i0}(f_k) \rvert$	Zero sequence current magnitude in p.u.
$i1(f_k) = \lvert \underline{i1}(f_k) \rvert$	Positive sequence current magnitude in p.u.
$i2(f_k) = \lvert \underline{i2}(f_k) \rvert$	Negative sequence current magnitude in p.u.

Monitor variables:

$irms{:}A = \sqrt{\sum_{k\geq 1} i{:}A(f_k)^2}$	Phase currents in p.u. based on rated current of element
$irms{:}B = \sqrt{\sum_{k\geq 1} i{:}B(f_k)^2}$	
$irms{:}C = \sqrt{\sum_{k\geq 1} i{:}C(f_k)^2}$	
$irms{:}N = \sqrt{\sum_{k\geq 1} i{:}N(f_k)^2}$	
$Irms{:}A = irms{:}A \cdot I_{nom}$	Phase currents in A
$Irms{:}B = irms{:}B \cdot I_{nom}$	
$Irms{:}C = irms{:}C \cdot I_{nom}$	
$Irms{:}N = irms{:}N \cdot I_{nom}$	
$HD{:}A = \frac{i{:}A(f_{out})}{i{:}A(f_1)} \cdot 100$	Harmonic distortion in %; based on fundamental frequency values
$HD{:}B = \frac{i{:}B(f_{out})}{i{:}B(f_1)} \cdot 100$	
$HD{:}C = \frac{i{:}C(f_{out})}{i{:}C(f_1)} \cdot 100$	
$HD{:}N = \frac{i{:}N(f_{out})}{i{:}N(f_1)} \cdot 100$	
$HD{:}A = \frac{i{:}A(f_{out})}{i_{nom}(f_1)} \cdot 100$	Harmonic distortion in %; based on nominal current
$HD{:}B = \frac{i{:}B(f_{out})}{i_{nom}(f_1)} \cdot 100$	
$HD{:}C = \frac{i{:}C(f_{out})}{i_{nom}(f_1)} \cdot 100$	
$HD{:}N = \frac{i{:}N(f_{out})}{i_{nom}(f_1)} \cdot 100$	
$HF{:}A = \frac{i{:}A(f_{out})}{irms{:}A(f_1)} \cdot 100$	Harmonic factor in %
$HF{:}B = \frac{i{:}B(f_{out})}{irms{:}B(f_1)} \cdot 100$	
$HF{:}C = \frac{i{:}C(f_{out})}{irms{:}C(f_1)} \cdot 100$	
$HF{:}N = \frac{i{:}N(f_{out})}{irms{:}N(f_1)} \cdot 100$	

$THD{:}A = \frac{1}{i{:}A(f_1)} \cdot \sqrt{irms{:}A^2 - i{:}A(f_1)^2} \cdot 100$

Total harmonic distortion, current in %; based on fundamental frequency values

$THD{:}B = \frac{1}{i{:}B(f_1)} \cdot \sqrt{irms{:}B^2 - i{:}B(f_1)^2} \cdot 100$

$THD{:}C = \frac{1}{i{:}C(f_1)} \cdot \sqrt{irms{:}C^2 - i{:}C(f_1)^2} \cdot 100$

$THD{:}N = \frac{1}{i{:}N(f_1)} \cdot \sqrt{irms{:}N^2 - i{:}N(f_1)^2} \cdot 100$

$THD{:}A = \frac{1}{i_{nom}} \cdot \sqrt{irms{:}A^2 - i{:}A(f_1)^2} \cdot 100$

Total harmonic distortion, current in %; based on nominal current

$THD{:}B = \frac{1}{i_{nom}} \cdot \sqrt{irms{:}B^2 - i{:}B(f_1)^2} \cdot 100$

$THD{:}C = \frac{1}{i_{nom}} \cdot \sqrt{irms{:}C^2 - i{:}C(f_1)^2} \cdot 100$

$THD{:}N = \frac{1}{i_{nom}} \cdot \sqrt{irms{:}N^2 - i{:}N(f_1)^2} \cdot 100$

$THF{:}A = \frac{1}{irms{:}A} \cdot \sqrt{irms{:}A^2 - i{:}A(f_1)^2} \cdot 100$

Total harmonic factor, current in %

$THF{:}B = \frac{1}{irms{:}B} \cdot \sqrt{irms{:}B^2 - i{:}B(f_1)^2} \cdot 100$

$THF{:}C = \frac{1}{irms{:}C} \cdot \sqrt{irms{:}C^2 - i{:}C(f_1)^2} \cdot 100$

$THF{:}N = \frac{1}{irms{:}N} \cdot \sqrt{irms{:}N^2 - i{:}N(f_1)^2} \cdot 100$

$TP{:}A = \sum_{k \geq 1} P{:}A(f_k)$

Total active power in MW

$TP{:}B = \sum_{k \geq 1} P{:}B(f_k)$

$TP{:}C = \sum_{k \geq 1} P{:}C(f_k)$

$TP{:}N = \sum_{k \geq 1} P{:}N(f_k)$

$TS{:}A = \sqrt{Irms{:}A^2 \cdot Urms{:}A^2}$

Total apparent power in MVA, where $Urms{:}A = \sqrt{\sum_{k \geq 1} |\underline{u}_a(f_k)|^2} \cdot U_{nom}$

$TS{:}B = \sqrt{Irms{:}B^2 \cdot Urms{:}B^2}$

$TS{:}C = \sqrt{Irms{:}C^2 \cdot Urms{:}C^2}$

$TS{:}N = \sqrt{Irms{:}N^2 \cdot Urms{:}N^2}$

$TQ{:}A = \sqrt{TS{:}A^2 - TP{:}A^2}$ Total reactive power in Mvar

$TQ{:}B = \sqrt{TS{:}B^2 - TP{:}B^2}$

$TQ{:}C = \sqrt{TS{:}C^2 - TP{:}C^2}$

$TQ{:}N = \sqrt{TS{:}N^2 - TP{:}N^2}$

$Tcosphi{:}A = \frac{TP{:}A}{TS{:}A}$ Total power factor

$Tcosphi{:}B = \frac{TP{:}B}{TS{:}B}$

$Tcosphi{:}C = \frac{TP{:}C}{TS{:}C}$

$Tcosphi{:}N = \frac{TP{:}N}{TS{:}N}$

$TAD{:}A = \frac{1}{i{:}A(f_1)} \cdot \left[\sum_{k\geq 1} i{:}A(f_k) - i{:}A(f_1)\right] \cdot 100$ Arithmetic distortion in %

$TAD{:}B = \frac{1}{i{:}B(f_1)} \cdot \left[\sum_{k\geq 1} i{:}B(f_k) - i{:}B(f_1)\right] \cdot 100$

$TAD{:}C = \frac{1}{i{:}C(f_1)} \cdot \left[\sum_{k\geq 1} i{:}C(f_k) - i{:}C(f_1)\right] \cdot 100$

$TAD{:}N = \frac{1}{i{:}N(f_1)} \cdot \left[\sum_{k\geq 1} i{:}N(f_k) - i{:}N(f_1)\right] \cdot 100$

$IT{:}A = \sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot i{:}A(f_k)^2} \cdot I_{nom}$ IT-product in kA

$IT{:}B = \sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot i{:}B(f_k)^2} \cdot I_{nom}$

$IT{:}C = \sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot i{:}C(f_k)^2} \cdot I_{nom}$

$IT{:}N = \sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot i{:}N(f_k)^2} \cdot I_{nom}$

$I0rms = \sqrt{\sum_{k\geq 1} i0(f_k)^2} \cdot I_{nom}$ Zero sequence current, RMS value in kA

$i0rms = \frac{I0rms}{I_{nom}}$ Zero sequence current, RMS value in p.u.

$THD0 = \frac{1}{i0(f_1)}\sqrt{i0rms^2 - i0(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, zero sequence); based on fundamental frequency values
$THD0 = \frac{1}{i_{nom}}\sqrt{i0rms^2 - i0(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, zero sequence); based on nominal current
$THF0 = \frac{1}{i0rms}\sqrt{i0rms^2 - i0(f_1)^2} \cdot 100$	Total harmonic factor in % (current, zero sequence)
$I1rms = \sqrt{\sum_{k\geq 1} i1(f_k)^2} \cdot I_{nom}$	Positive sequence current, RMS value in kA
$i1rms = \frac{I1rms}{I_{nom}}$	Positive sequence current, RMS value in p.u.
$THD1 = \frac{1}{i1(f_1)}\sqrt{i1rms^2 - i1(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, positive sequence); based on fundamental frequency values
$THD1 = \frac{1}{i_{nom}}\sqrt{i1rms^2 - i1(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, positive sequence); based on nominal current
$THF1 = \frac{1}{i1rms}\sqrt{i1rms^2 - i1(f_1)^2} \cdot 100$	Total harmonic factor in % (current, positive sequence)
$I2rms = \sqrt{\sum_{k\geq 1} i2(f_k)^2} \cdot I_{nom}$	Negative sequence current, RMS value in kA
$i2rms = \frac{I2rms}{I_{nom}}$	Negative sequence current, RMS value in p.u.
$THD2 = \frac{1}{i2(f_1)}\sqrt{i2rms^2 - i2(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, negative sequence); based on fundamental frequency values
$THD2 = \frac{1}{i_{nom}}\sqrt{i2rms^2 - i2(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, negative sequence); based on nominal current
$THF2 = \frac{1}{i2rms}\sqrt{i2rms^2 - i2(f_1)^2} \cdot 100$	Total harmonic factor in % (current, negative sequence)
$HD0 = \frac{i0(f_{out})}{\sqrt{i0(f_1)^2 + i1(f_1)^2 + i2(f_1)^2}} \cdot 100$	Harmonic distortion, zero sequence in %; based on fundamental frequency values
$HD1 = \frac{i1(f_{out})}{\sqrt{i0(f_1)^2 + i1(f_1)^2 + i2(f_1)^2}} \cdot 100$	Harmonic distortion, positive sequence in %
$HD2 = \frac{i2(f_{out})}{\sqrt{i0(f_1)^2 + i1(f_1)^2 + i2(f_1)^2}} \cdot 100$	Harmonic distortion, negative sequence in %

$HD0 = \frac{i0(f_{out})}{i_{nom}} \cdot 100$	Harmonic distortion, zero sequence in %; based on nominal current
$HD1 = \frac{i1(f_{out})}{i_{nom}} \cdot 100$	Harmonic distortion, positive sequence in %
$HD2 = \frac{i2(f_{out})}{i_{nom}} \cdot 100$	Harmonic distortion, negative sequence in %
$HF0 = \frac{i0(f_{out})}{i0rms} \cdot 100$	Harmonic factor, zero sequence in %
$HF1 = \frac{i1(f_{out})}{i1rms} \cdot 100$	Harmonic factor, positive sequence in %
$HF2 = \frac{i2(f_{out})}{i2rms} \cdot 100$	Harmonic factor, negative sequence in %
$TPsum = \sum_{x=[A,B,C,N]} TP{:}x$	Total active power in MW
$TQsum = \sum_{x=[A,B,C,N]} TQ{:}x$	Total reactive power in Mvar
$TSsum = \sqrt{TPsum^2 + TQsum^2}$	Total apparent power in MVA
$Tcosphisum = \frac{TPsum}{TSsum}$	Total power factor
$THDbal = \sqrt{THD1^2 + THD2^2}$	Total harmonic distortion in %, excluding zero sequence
$THDtot = \sqrt{THD1^2 + THD2^2 + THD0^2}$	Total harmonic distortion in %, including zero sequence
$TADmx = \max_{x=[A,B,C,N]}(TAD{:}x)$	Arithmetic distortion in %
$TADint = \max_{x=[A,B,C,N]}(TADint{:}x)$	Arithmetic distortion in %, where $TADint{:}x = \frac{1}{i{:}x(f_1)} \cdot \left[\sum_{\substack{k \geq 1 \\ integer}} i{:}x(f_k) - i{:}x(f_1) \right] \cdot 100$
$TADnint = \max_{x=[A,B,C,N]}(TADnint{:}x)$	Arithmetic distortion in %, where $TADnint{:}x = \frac{1}{i{:}x(f_1)} \cdot \left[\sum_{\substack{k \geq 1 \\ non-integer}} i{:}x(f_k) - i{:}x(f_1) \right] \cdot 100$
$TAD_3 = \max_{x=[A,B,C,N]}(TADint{:}x)$	Arithmetic distortion in %, harmonic orders: multiples of three
$TAD_2 = \max_{x=[A,B,C,N]}(TADint{:}x)$	Arithmetic distortion in %, harmonic orders: multiples of two

$THDint = \max_{x=[A,B,C,N]}(THDint{:}x)$	Total harmonic distortion in %, where $THDint{:}x = \frac{1}{i{:}x(f_1)} \cdot \sqrt{irms{:}x^2 - i{:}x(f_1)^2} \cdot 100$ (Fundamental frequency values) or $THDint{:}x = \frac{1}{inom} \cdot \sqrt{irms{:}x^2 - i{:}x(f_1)^2} \cdot 100$ (Nominal current) and $irms{:}x = \sqrt{\sum_{\substack{k\geq 1 \\ integer}} i{:}x(f_k)^2}$
$THDnint = \max_{x=[A,B,C,N]}(THDnint{:}x)$	Total harmonic distortion in %, where $THDnint{:}x = \frac{1}{i{:}x(f_1)} \cdot \sqrt{irms{:}x^2 - i{:}x(f_1)^2} \cdot 100$ (Fundamental frequency values) or $THDnint{:}x = \frac{1}{inom} \cdot \sqrt{irms{:}x^2 - i{:}x(f_1)^2} \cdot 100$ (Nominal current) and $irms{:}x = \sqrt{\sum_{\substack{k>1 \\ non-integer}} i{:}x(f_k)^2}$
$THD_3 = \max_{x=[A,B,C,N]}(THDint{:}x)$	Total harmonic distortion in %, harmonic orders: multiples of three
$THD_2 = \max_{x=[A,B,C,N]}(THDint{:}x)$	Total harmonic distortion in %, harmonic orders: multiples of two
$ITmx = \max(IT{:}x)$	IT-Product in kA

Calculation variables: The following calculation variables are available for lines, transformers and series compensation:

$LossesTot = \sum_{h\geq 1}^{H} \sum_{x=[A,B,C,N]} Losses{:}x$	Total losses, including the fundamental in kW, where h is the harmonic order
$LossesHrm = \sum_{h>1}^{H} \sum_{x=[A,B,C,N]} Losses{:}x$	Harmonic losses in kW
$loadingTot = irms_{max}/i_{nom} \cdot 100$	Total loading in %

2.2.0.2 Bus Monitor Variables

Table2.4 provides the basic nomenclature used throughout this section.

Table 2.4: Nomenclature

Name	Unit	Description
f_k	Hz	Frequency of harmonic order k
f_{out}	Hz	Output frequency specified by user in Harmonic Load Flow Command (parameter name: *ifshow*)
$\underline{u}_a(f_k)$	p.u.	Complex harmonic voltage, phase A. Similar nomenclature is used for the other phases.
$\underline{ul}_a(f_k)$	p.u.	Complex line-line voltage, phase A, where $\underline{ul}_a(f_k) = \underline{u}_a(f_k) - \underline{u}_b(f_k)$.
$ul{:}A(f_k)$	p.u.	Line-line voltage magnitude, phase A, where $ul{:}A(f_k) = \lvert\underline{u}_a(f_k) - \underline{u}_b(f_k)\rvert$. Similar follows for the other phases.
$\underline{uln}_a(f_k)$	p.u.	Complex line-neutral voltage, phase A, where $\underline{uln}_a(f_k) = \underline{u}_a(f_k) - \underline{u}_n(f_k)$. Similar follows for the other phases.
$uln{:}A(f_k)$	p.u.	Line-line voltage magnitude, phase A, where $uln{:}A(f_k) = \lvert\underline{u}_a(f_k) - \underline{u}_n(f_k)\rvert$. Similar follows for the other phases.
$u{:}A(f_k)$	p.u.	Voltage magnitude at frequency of harmonic order k
U_{nom}	kV	Nominal voltage of the busbar in kV
$\phi_{initial_a}$	deg	Initial voltage angle, phase A. Similar nomenclature is used for other phases.
$\underline{Zl}_a$	Ω	Line impedance, phase A. Similar follows for the other phases.
$\underline{Zln}_a$	Ω	Line-neutral impedance, phase A. Similar follows for the other phases.
$\underline{Z}_a$	Ω	Short-circuit impedance, phase A. Similar nomenclature is used for other phases.
$\underline{Z}_{nc,a}$	Ω	Short-circuit impedance without capacitive effects, phase A. Similar nomenclature is used for other phases (only available for the Frequency Sweep calculation).

Monitor Variables for Output Frequency The following monitor variables are calculated for user-selected output frequency, f_{out}:

Intermediate variables:

$\underline{u0} = \frac{1}{3}\left(\underline{u}_a + \underline{u}_b + \underline{u}_c\right)$	Zero sequence voltage in p.u.
$\underline{u1} = \frac{1}{3}\left(\underline{u}_a + a \cdot \underline{u}_b + a^2 \cdot \underline{u}_c\right)$	Positive sequence voltage in p.u.
$\underline{u2} = \frac{1}{3}\left(\underline{u}_a + a^2 \cdot \underline{u}_b + a \cdot \underline{u}_c\right)$	Negative sequence voltage in p.u.

Monitor variables:

$u{:}A = \lvert\underline{u}_a\rvert$	Line-ground voltage, magnitude in p.u.
$u{:}B = \lvert\underline{u}_b\rvert$	
$u{:}C = \lvert\underline{u}_c\rvert$	
$upc{:}A = u{:}A \cdot 100$	Line-ground voltage, magnitude in %
$upc{:}B = u{:}B \cdot 100$	
$upc{:}C = u{:}C \cdot 100$	
$U{:}A = u{:}A \cdot U_{nom}/\sqrt{3}$	Line-ground voltage magnitude in kV
$U{:}B = u{:}B \cdot U_{nom}/\sqrt{3}$	
$U{:}C = u{:}C \cdot U_{nom}/\sqrt{3}$	
$phiu{:}A = \phi_{\underline{u}_a}$	Voltage angle in deg
$phiu{:}B = \phi_{\underline{u}_b}$	
$phiu{:}C = \phi_{\underline{u}_c}$	
$phiurel{:}A = \phi_{\underline{u}_a} - \phi_{initial_a}$	Voltage, relative angle in deg
$phiurel{:}B = \phi_{\underline{u}_b} - \phi_{initial_b}$	
$phiurel{:}C = \phi_{\underline{u}_c} - \phi_{initial_c}$	
$ul{:}A = \lvert\underline{ul}_a\rvert$	Line-line voltage magnitude in p.u.
$ul{:}B = \lvert\underline{ul}_b\rvert$	
$ul{:}C = \lvert\underline{ul}_c\rvert$	
$ulpc{:}A = ul{:}A \cdot 100$	Line-line voltage magnitude in %
$ulpc{:}B = ul{:}B \cdot 100$	
$ulpc{:}C = ul{:}C \cdot 100$	

$Ul{:}A = ul{:}A \cdot U_{nom}$	Line-line voltage magnitude in kV
$Ul{:}B = ul{:}B \cdot U_{nom}$	
$Ul{:}C = ul{:}C \cdot U_{nom}$	
$phiul{:}A = \phi_{\underline{ul}_a}$	Line-line voltage angle in deg
$phiul{:}B = \phi_{\underline{ul}_b}$	
$phiul{:}C = \phi_{\underline{ul}_c}$	
$uln{:}A = \lvert\underline{uln}_a\rvert$	Line-neutral voltage magnitude in p.u.
$uln{:}B = \lvert\underline{uln}_b\rvert$	
$uln{:}C = \lvert\underline{uln}_c\rvert$	
$Uln{:}A = uln{:}A \cdot U_{nom}/\sqrt{3}$	Line-neutral voltage magnitude in kV
$Uln{:}B = uln{:}B \cdot U_{nom}/\sqrt{3}$	
$Uln{:}C = uln{:}C \cdot U_{nom}/\sqrt{3}$	
$phiuln{:}A = \phi_{\underline{uln}_a}$	Line-neutral voltage angle in deg
$phiuln{:}B = \phi_{\underline{uln}_b}$	
$phiuln{:}C = \phi_{\underline{uln}_c}$	
$R{:}A = \Re(\underline{Z}_a)$	Line-ground resistance in Ω
$R{:}B = \Re(\underline{Z}_b)$	
$R{:}C = \Re(\underline{Z}_c)$	
$X{:}A = \Im(\underline{Z}_a)$	Line-ground network reactance in Ω
$X{:}B = \Im(\underline{Z}_b)$	
$X{:}C = \Im(\underline{Z}_c)$	
$Z{:}A = \lvert\underline{Z}_a\rvert$	Line-ground network impedance in Ω
$Z{:}B = \lvert\underline{Z}_b\rvert$	
$Z{:}C = \lvert\underline{Z}_c\rvert$	

$phiz{:}A = \phi_{\underline{Z}_a}$	Line-ground angle of network impedance in deg
$phiz{:}B = \phi_{\underline{Z}_b}$	
$phiz{:}C = \phi_{\underline{Z}_c}$	
$Rl{:}A = \Re(\underline{Z}l_a)$	Line-line resistance in Ω
$Rl{:}B = \Re(\underline{Z}l_b)$	
$Rl{:}C = \Re(\underline{Z}l_c)$	
$Xl{:}A = \Im(\underline{Z}l_a)$	Line-line network reactance in Ω
$Xl{:}B = \Im(\underline{Z}l_b)$	
$Xl{:}C = \Im(\underline{Z}l_c)$	
$Zl{:}A = \lvert\underline{Z}l_a\rvert$	Line-line network impedance in Ω
$Zl{:}B = \lvert\underline{Z}l_b\rvert$	
$Zl{:}C = \lvert\underline{Z}l_c\rvert$	
$phizl{:}A = \phi_{\underline{Z}l_a}$	Line-line angle of network impedance in deg
$phizl{:}B = \phi_{\underline{Z}l_b}$	
$phizl{:}C = \phi_{\underline{Z}l_c}$	
$Rln{:}A = \Re(\underline{Z}ln_a)$	Line-neutral resistance in Ω
$Rln{:}B = \Re(\underline{Z}ln_b)$	
$Rln{:}C = \Re(\underline{Z}ln_c)$	
$Xln{:}A = \Im(\underline{Z}ln_a)$	Line-neutral network reactance in Ω
$Xln{:}B = \Im(\underline{Z}ln_b)$	
$Xln{:}C = \Im(\underline{Z}ln_c)$	
$Zln{:}A = \lvert\underline{Z}ln_a\rvert$	Line-neutral network impedance in Ω
$Zln{:}B = \lvert\underline{Z}ln_b\rvert$	
$Zln{:}C = \lvert\underline{Z}ln_c\rvert$	

$phizln{:}A = \phi_{\underline{Zln}_a}$	Line-neutral angle of network impedance in deg
$phizln{:}B = \phi_{\underline{Zln}_b}$	
$phizln{:}C = \phi_{\underline{Zln}_c}$	
$u0 = \lvert\underline{u0}\rvert$	Line-ground zero sequence voltage magnitude in p.u.
$U0 = u0 \cdot U_{nom}/\sqrt{3}$	Zero sequence voltage magnitude in kV
$U0x3 = 3 \cdot U0$	in kV
$phiu0 = \phi_{\underline{u0}}$	Zero sequence voltage angle in deg
$u1 = \lvert\underline{u1}\rvert$	Line-ground positive sequence voltage magnitude in p.u.
$u1pc = u1 \cdot 100$	Line-ground positive sequence voltage magnitude in %
$u1r = \Re(\underline{u1})$	Positive sequence voltage magnitude, real part in p.u.
$u1i = \Im(\underline{u1})$	Positive sequence voltage magnitude, imaginary part in p.u.
$U1 = u1 \cdot U_{nom}/\sqrt{3}$	Line-ground positive sequence voltage magnitude in kV
$phiu1 = \phi_{\underline{u1}}$	Line-ground positive sequence voltage angle in deg
$u2 = \lvert\underline{u2}\rvert$	Line-ground negative sequence voltage magnitude in p.u.
$U2 = u2 \cdot U_{nom}/\sqrt{3}$	Line-ground negative sequence voltage magnitude in kV
$phiu2 = \phi_{\underline{u2}}$	Line-ground negative sequence voltage magnitude in deg
$U1l = u1 \cdot U_{nom}$	Line-line positive sequence voltage magnitude in kV
$U2l = u2 \cdot U_{nom}$	Line-line negative sequence voltage magnitude in kV
$un = \lvert\underline{u}_n\rvert$	Neutral-ground voltage magnitude in p.u.
$Un = un \cdot U_{nom}/\sqrt{3}$	Neutral-ground voltage magnitude in kV
$R0 = \Re(\underline{Z0})$	Zero sequence resistance in Ω
$X0 = \Im(\underline{Z0})$	Zero sequence network reactance in Ω
$Z0 = \lvert\underline{Z0}\rvert$	Zero sequence network impedance in Ω
$phiz0 = \phi_{\underline{Z0}}$	Zero sequence network impedance angle in deg

$R1 = \Re(\underline{Z1})$	Positive sequence resistance in Ω
$X1 = \Im(\underline{Z1})$	Positive sequence network reactance in Ω
$Z1 = \lvert\underline{Z1}\rvert$	Positive sequence network impedance in Ω
$phiz1 = \phi_{\underline{Z1}}$	Positive sequence network impedance angle in deg
$R2 = \Re(\underline{Z2})$	Negative sequence resistance in Ω
$X2 = \Im(\underline{Z2})$	Negative sequence network reactance in Ω
$Z2 = \lvert\underline{Z2}\rvert$	Negative sequence network impedance in Ω
$phiz2 = \phi_{\underline{Z2}}$	Negative sequence network impedance angle in deg
$Rn = \Re(\underline{Z}_n)$	Resistance, neutral in Ω
$Xn = \Im(\underline{Z}_n)$	Network reactance, neutral in Ω
$Zn = \lvert\underline{Z}_n\rvert$	Network impedance, neutral in Ω
$phizn = \phi_{\underline{Z}_n}$	Angle of network impedance, neutral in Ω
$ubfac = \frac{u2}{u1}$	Unbalance factor

and the following quantities for the short-circuit impedance without capacitive effects ($Z_{nc,a/b/c}$). The quantities are only available for the Frequency Sweep calculation when one of the following calculation variables is selected for recording, otherwise the value is set to zero.

$Rnc{:}A = \Re(\underline{Z}_{nc,a})$	Line-ground resistance (without C) in Ω
$Rnc{:}B = \Re(\underline{Z}_{nc,b})$	
$Rnc{:}C = \Re(\underline{Z}_{nc,c})$	
$Xnc{:}A = \Im(\underline{Z}_{nc,a})$	Line-ground network reactance (without C) in Ω
$Xnc{:}B = \Im(\underline{Z}_{nc,b})$	
$Xnc{:}C = \Im(\underline{Z}_{nc,c})$	
$Znc{:}A = \lvert\underline{Z}_{nc,a}\rvert$	Line-ground network (without C) impedance in Ω
$Znc{:}B = \lvert\underline{Z}_{nc,b}\rvert$	
$Znc{:}C = \lvert\underline{Z}_{nc,c}\rvert$	

$phiznc{:}A = \phi_{\underline{Z}_{nc,a}}$	Line-ground angle of network impedance (without C) in deg
$phiznc{:}B = \phi_{\underline{Z}_{nc,b}}$	
$phiznc{:}C = \phi_{\underline{Z}_{nc,c}}$	
$R0nc = \Re(\underline{Z0}_{nc})$	Zero sequence resistance (without C) in Ω
$X0nc = \Im(\underline{Z0}_{nc})$	Zero sequence network reactance (without C) in Ω
$Z0nc = \lvert\underline{Z0}_{nc}\rvert$	Zero sequence network impedance (without C) in Ω
$phiz0nc = \phi_{\underline{Z0}_{nc}}$	Zero sequence network impedance angle (without C) in deg
$R1nc = \Re(\underline{Z1}_{nc})$	Positive sequence resistance (without C) in Ω
$X1nc = \Im(\underline{Z1}_{nc})$	Positive sequence network reactance (without C) in Ω
$Z1nc = \lvert\underline{Z1}_{nc}\rvert$	Positive sequence network impedance (without C) in Ω
$phiz1nc = \phi_{\underline{Z1}_{nc}}$	Positive sequence network impedance angle (without C) in deg
$R2nc = \Re(\underline{Z2}_{nc})$	Negative sequence resistance (without C) in Ω
$X2nc = \Im(\underline{Z2}_{nc})$	Negative sequence network reactance (without C) in Ω
$Z2nc = \lvert\underline{Z2}_{nc}\rvert$	Negative sequence network impedance (without C) in Ω
$phiz2nc = \phi_{\underline{Z2}_{nc}}$	Negative sequence network impedance angle (without C) in deg

Monitor Variables Considering All Frequencies: The following monitor variables are calculated considering all frequencies, f_k:

Intermediate variables:

$\underline{u0}(f_k) = \frac{1}{3}\left[\underline{u}_a(f_k) + \underline{u}_b(f_k) + \underline{u}_c(f_k)\right]$	Zero sequence voltage in p.u.
$\underline{u1}(f_k) = \frac{1}{3}\left[\underline{u}_a(f_k) + a \cdot \underline{u}_b(f_k) + a^2 \cdot \underline{u}_c(f_k)\right]$	Positive sequence voltage in p.u.
$\underline{u2}(f_k) = \frac{1}{3}\left[\underline{u}_a(f_k) + a^2 \cdot \underline{u}_b(f_k) + a \cdot \underline{u}_c(f_k)\right]$	Negative sequence voltage in p.u.
$u0(f_k) = \lvert\underline{u0}(f_k)\rvert$	Zero sequence voltage magnitude in p.u.
$u1(f_k) = \lvert\underline{u1}(f_k)\rvert$	Positive sequence voltage magnitude in p.u.
$u2(f_k) = \lvert\underline{u2}(f_k)\rvert$	Negative sequence voltage magnitude in p.u.

$u{:}A(f_k) = \lvert\underline{u}_a(f_k)\rvert$	Line-ground voltage, magnitude in p.u.
$u{:}B(f_k) = \lvert\underline{u}_b(f_k)\rvert$	
$u{:}C(f_k) = \lvert\underline{u}_c(f_k)\rvert$	

$ul{:}A(f_k) = \lvert\underline{ul}_a(f_k)\rvert$	Line-line voltage magnitude in p.u.
$ul{:}C(f_k) = \lvert\underline{ul}_c(f_k)\rvert$	
$ul{:}B(f_k) = \lvert\underline{ul}_b(f_k)\rvert$	

Monitor variables:

$urms{:}A = \sqrt{\sum_{k\geq 1} u{:}A(f_k)^2}$	RMS value of line-neutral voltage in p.u.
$urms{:}B = \sqrt{\sum_{k\geq 1} u{:}B(f_k)^2}$	
$urms{:}C = \sqrt{\sum_{k\geq 1} u{:}C(f_k)^2}$	
$Urms{:}A = urms{:}A \cdot U_{nom}/\sqrt{3}$	RMS value of line-neutral voltage in kV
$Urms{:}B = urms{:}B \cdot U_{nom}/\sqrt{3}$	
$Urms{:}C = urms{:}C \cdot U_{nom}/\sqrt{3}$	
$u0rms = \sqrt{\sum_{k\geq 1} u0(f_k)^2}$	RMS value of zero sequence voltage in p.u.
$U0rms = u0rms \cdot U_{nom}/\sqrt{3}$	RMS value of zero sequence voltage in kV
$u1rms = \sqrt{\sum_{k\geq 1} u1(f_k)^2}$	RMS value of positive sequence voltage in p.u.
$U1rms = u1rms \cdot U_{nom}/\sqrt{3}$	RMS value of positive sequence voltage in kV
$U1lrms = u1rms \cdot U_{nom}$	RMS value of positive sequence line-line voltage in kV
$u2rms = \sqrt{\sum_{k\geq 1} u2(f_k)^2}$	RMS value of negative sequence voltage in p.u.
$U2rms = u2rms \cdot U_{nom}/\sqrt{3}$	RMS value of negative sequence voltage in kV
$U2lrms = u2rms \cdot U_{nom}$	RMS value of negative sequence line-line voltage in kV
$HD{:}A = \frac{u{:}A(f_{out})}{u{:}A(f_1)} \cdot 100$	Harmonic distortion in %; based on fundamental frequency values
$HD{:}B = \frac{u{:}B(f_{out})}{u{:}B(f_1)} \cdot 100$	
$HD{:}C = \frac{u{:}C(f_{out})}{u{:}C(f_1)} \cdot 100$	
$HD{:}A = \frac{u{:}A(f_{out})}{u_{nom}{:}A} \cdot 100$	Harmonic distortion in %; based on nominal voltage, where $u_{nom} = 1$ p.u.
$HD{:}B = \frac{u{:}B(f_{out})}{u_{nom}{:}B} \cdot 100$	
$HD{:}C = \frac{u{:}C(f_{out})}{u_{nom}{:}C} \cdot 100$	

$HF{:}A = \frac{u{:}A(f_{out})}{urms{:}A} \cdot 100$ Harmonic factor in %

$HF{:}B = \frac{u{:}B(f_{out})}{urms{:}B} \cdot 100$

$HF{:}C = \frac{u{:}C(f_{out})}{urms{:}C} \cdot 100$

$HD0 = \frac{u0(f_{out})}{\sqrt{u0(f_1)^2+u1(f_1)^2+u2(f_1)^2}} \cdot 100$ Harmonic distortion, zero sequence in %; based on fundamental frequency values

$HD1 = \frac{u1(f_{out})}{\sqrt{u0(f_1)^2+u1(f_1)^2+u2(f_1)^2}} \cdot 100$ Harmonic distortion, positive sequence in %

$HD2 = \frac{u2(f_{out})}{\sqrt{u0(f_1)^2+u1(f_1)^2+u2(f_1)^2}} \cdot 100$ Harmonic distortion, negative sequence in %

$HF0 = \frac{u0(f_{out})}{u0rms} \cdot 100$ Harmonic factor, zero sequence in %

$HF1 = \frac{u1(f_{out})}{u1rms} \cdot 100$ Harmonic factor, positive sequence in %

$HF2 = \frac{u2(f_{out})}{u2rms} \cdot 100$ Harmonic factor, negative sequence in %

$HD0 = \frac{u0(f_{out})}{u_{nom}} \cdot 100$ Harmonic distortion, zero sequence in %; based on nominal voltage, where $u_{nom} = 1$ p.u.

$HD1 = \frac{u1(f_{out})}{u_{nom}} \cdot 100$ Harmonic distortion, positive sequence in %

$HD2 = \frac{u2(f_{out})}{u_{nom}} \cdot 100$ Harmonic distortion, negative sequence in %

$THD{:}A = \frac{1}{u{:}A(f_1)} \sqrt{urms{:}A^2 - u{:}A(f_1)^2} \cdot 100$ Total harmonic distortion in %; based on fundamental frequency values

$THD{:}B = \frac{1}{u{:}B(f_1)} \sqrt{urms{:}B^2 - u{:}B(f_1)^2} \cdot 100$

$THD{:}C = \frac{1}{u{:}C(f_1)} \sqrt{urms{:}C^2 - u{:}C(f_1)^2} \cdot 100$

$THD{:}A = \frac{1}{u_{nom}{:}A} \sqrt{urms{:}A^2 - u{:}A(f_1)^2} \cdot 100$ Total harmonic distortion in %; based on nominal voltage, where $u_{nom} = 1$ p.u.

$THD{:}B = \frac{1}{u_{nom}{:}B} \sqrt{urms{:}B^2 - u{:}B(f_1)^2} \cdot 100$

$THD{:}C = \frac{1}{u_{nom}{:}C} \sqrt{urms{:}C^2 - u{:}C(f_1)^2} \cdot 100$

$THF{:}A = \frac{1}{urms{:}A}\sqrt{urms{:}A^2 - u{:}A(f_1)^2} \cdot 100$ Total harmonic factor in %

$THF{:}B = \frac{1}{urms{:}B}\sqrt{urms{:}B^2 - u{:}B(f_1)^2} \cdot 100$

$THF{:}C = \frac{1}{urms{:}C}\sqrt{urms{:}C^2 - u{:}C(f_1)^2} \cdot 100$

$ulrms{:}A = \sqrt{\sum_{k\geq 1} ul{:}A(f_k)^2}$ RMS value of line-line voltage in p.u.

$ulrms{:}B = \sqrt{\sum_{k\geq 1} ul{:}B(f_k)^2}$

$ulrms{:}C = \sqrt{\sum_{k\geq 1} ul{:}C(f_k)^2}$

$Ulrms{:}A = ulrms{:}A \cdot U_{nom}$ RMS value of line-line voltage in kV

$Ulrms{:}B = ulrms{:}B \cdot U_{nom}$

$Ulrms{:}C = ulrms{:}C \cdot U_{nom}$

$TAD{:}A = \frac{1}{u{:}A(f_1)} \cdot \left[\sum_{k\geq 1} u{:}A(f_k) - u{:}A(f_1)\right] \cdot 100$ Arithmetic distortion in %

$TAD{:}B = \frac{1}{u{:}B(f_1)} \cdot \left[\sum_{k\geq 1} u{:}B(f_k) - u{:}B(f_1)\right] \cdot 100$

$TAD{:}C = \frac{1}{u{:}C(f_1)} \cdot \left[\sum_{k\geq 1} u{:}C(f_k) - u{:}C(f_1)\right] \cdot 100$

$uasum{:}A = \sum_{k\geq 1} u{:}A(f_k)$ Arithmetic voltage sum in p.u.

$uasum{:}B = \sum_{k\geq 1} u{:}B(f_k)$

$uasum{:}C = \sum_{k\geq 1} u{:}C(f_k)$

$TIF{:}A = \frac{1}{urms{:}A}\sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot u{:}A(f_k)^2}$ Telephone interference factor

$TIF{:}B = \frac{1}{urms{:}B}\sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot u{:}B(f_k)^2}$

$TIF{:}C = \frac{1}{urms{:}C}\sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot u{:}C(f_k)^2}$

$urmsint = \max_{x=[A,B,C]}(urmsint : x)$ RMS value of integer harmonics in p.u., where

$urmsint : x = \sqrt{\sum_{\substack{k\geq 1 \\ integer}} u : x(f_k)^2}$

$urmsnint = \max_{x=[A,B,C]}(urmsnint : x)$ RMS value of non-integer harmonics in p.u., where

$urmsnint : x = \sqrt{\sum_{\substack{k\geq 1 \\ non-integer}} u : x(f_k)^2}$

$THD0 = \frac{1}{u1(f_1)^2+u0(f_1)^2+u2(f_1)^2}\sqrt{u0rms^2 - u0(f_1)^2}$	Total harmonic distortion in % (voltage, zero sequence)
$THD1 = \frac{1}{u1(f_1)^2+u0(f_1)^2+u2(f_1)^2}\sqrt{u1rms^2 - u1(f_1)^2}$	Total harmonic distortion in % (voltage, positive sequence)
$THD2 = \frac{1}{u1(f_1)^2+u0(f_1)^2+u2(f_1)^2}\sqrt{u2rms^2 - u2(f_1)^2}$	Total harmonic distortion in % (voltage, negative sequence)
$THDbal = \sqrt{THD1^2 + THD2^2}$	Total harmonic distortion in %, excluding zero sequence
$THDtot = \sqrt{THD1^2 + THD2^2 + THD0^2}$	Total harmonic distortion in %, including zero sequence
$TADmx = \max_{x=[A,B,C]}(TAD{:}x)$	Arithmetic distortion in %

$TADint = \max_{x=[A,B,C]}(TADint{:}x)$

Arithmetic distortion in %, where

$$TADint{:}x = \frac{1}{u{:}x(f_1)} \cdot \left[\sum_{\substack{k\geq 1 \\ integer}} u{:}x(f_k) - u{:}x(f_1) \right] \cdot 100$$

$TADnint = \max_{x=[A,B,C]}(TADnint{:}x)$

Arithmetic distortion in %, where

$$TADnint{:}x = \frac{1}{u{:}x(f_1)} \cdot \left[\sum_{\substack{k>1 \\ non-integer}} u{:}x(f_k) - u{:}x(f_1) \right] \cdot 100$$

$TAD_3 = \max_{x=[A,B,C]}(TADint{:}x)$

Arithmetic distortion in %, harmonic orders: multiples of three

$TAD_2 = \max_{x=[A,B,C]}(TADint{:}x)$

Arithmetic distortion in %, harmonic orders: multiples of two

$THDint = \max_{x=[A,B,C]}(THDint{:}x)$

Total harmonic distortion in %, where

$$THDint{:}x = \frac{1}{u{:}x(f_1)} \cdot \sqrt{urms{:}x^2 - u{:}x(f_1)^2} \cdot 100$$

(Fundamental frequency values) or

$$THDint{:}x = \frac{1}{u_{nom}{:}x} \cdot \sqrt{urms{:}x^2 - u{:}x(f_1)^2} \cdot 100$$

(Nominal voltage) and

$$urms{:}x = \sqrt{\sum_{\substack{k\geq 1 \\ integer}} u{:}x(f_k)^2}$$

$THDnint = \max_{x=[A,B,C]}(THDnint{:}x)$

Total harmonic distortion in %, where

$$THDnint{:}x = \frac{1}{u{:}x(f_1)} \cdot \sqrt{urms{:}x^2 - u{:}x(f_1)^2} \cdot 100$$

(Fundamental frequency values) or

$$THDnint{:}x = \frac{1}{urms{:}x} \cdot \sqrt{urms{:}x^2 - u{:}x(f_1)^2} \cdot 100$$

(Nominal voltage) and

$$urms{:}x = \sqrt{\sum_{\substack{k>1 \\ non-integer}} u{:}x(f_k)^2}$$

$THD_3 = \max_{x=[A,B,C]}(THDint{:}x)$

Total harmonic distortion in %, harmonic orders: multiples of three

$THD_2 = \max_{x=[A,B,C]}(TADint{:}x)$

Total harmonic distortion in %, harmonic orders: multiples of two

$TIFmx = \max_{x=[A,B,C]}(TIF{:}x)$	Total interference factor, for harmonic orders
$uasummx = \max_{x=[A,B,C]}(uasum{:}x)$	Arithmetic voltage sum in p.u.
$uasumint = \max_{x=[A,B,C]}(uasumint{:}x)$	Arithmetic sum of integer harmonics in p.u., where $uasumint{:}x = \sum_{\substack{k\geq 2 \\ integer}} u{:}x(f_k)$
$uasumnint = \max_{x=[A,B,C]}(uasumnint{:}x)$	Arithmetic sum of integer harmonics in p.u., where $uasumnint{:}x = \sum_{\substack{k>2 \\ non-integer}} u{:}x(f_k)$

3 Calculation Considering IEC Harmonic Sources

3.1 Balanced Calculation

3.1.0.1 Branch Monitor Variables

Table 3.1 provides the basic nomenclature used throughout this section.

Table 3.1: Nomenclature

Name	Unit	Description
f_k		Frequency of harmonic order k
f_{out}		Output frequency specified by user in Harmonic Load Flow Command (parameter name: *ifshow*)
$\underline{i}(f_k)$		Complex harmonic current at frequency of harmonic order k (all p.u. values are based on I_{nom})
$\underline{u}(f_k)$		Complex harmonic voltage at frequency of harmonic order k (all p.u. values are based on U_{nom})
$\underline{I}(f_k)$		Complex harmonic current in A, at frequency of harmonic order k
$\underline{U}(f_k)$		Complex harmonic voltage in kV, at frequency of harmonic order k
I_{nom}, i_{nom}		Nominal current of element in A, in p.u.
U_{nom}, u_{nom}		Rated voltage of element in kV, in p.u.
$tif(f_k)$		Telephone influence factor at frequency of harmonic order k

PowerFactory calculates monitor variables for the Harmonic Analysis command (balanced calculation) in the following manner provided that at least one harmonic source in the network is

defined as 'IEC' (TypHmccur parameter name: *i_usym*).

Per IEC source (subscript $m \geq 1$) and per frequency (f_k), the quantities shown in Table 3.2 are available for the calculation of branch monitor variables:

Table 3.2: Nomenclature

Complex voltage	Complex current
$\underline{u}_m(f_k)$	$\underline{i}_m(f_k)$

For non-IEC sources (subscript $m = 0$), the quantities shown in Table 3.3 are available as a lumped representation of all non-IEC sources, for the calculation of branch monitor variables:

Table 3.3: Nomenclature

Complex voltage	Complex current
$\underline{u}_0(f_k)$	$\underline{i}_0(f_k)$

Monitor Variables for Output Frequency The following monitor variables are calculated for user-selected output frequency, f_{out}:

Intermediate variables:

$u = \sqrt[\alpha]{\sum_{m=0}^{N} \lvert\underline{u}_m\rvert^{\alpha}}$	Voltage magnitude in kV according to IEC-61000-3-6

Monitor variables:

$i = \sqrt[\alpha]{\sum_{m=0}^{N} \lvert\underline{i}_m\rvert^{\alpha}}$	Current magnitude in p.u., according to IEC-61000-3-6
$I = i \cdot I_{nom}$	Current magnitude in A
$I1 = I$	Positive sequence current magnitude in A
$I1 = 0$	Positive sequence current magnitude in A, for harmonic orders 5, 11, 17, 23,. . . ,
$I2 = I$	Negative sequence current magnitude in A, for harmonic orders 5, 11, 17, 23,. . . ,
$I2 = 0$	Negative sequence current magnitude in A
$phiui$	(N/A) Angle between voltage and current in deg
$phii$	(N/A) Current angle, absolute in deg

$P = ku \cdot ki \cdot \sum_{m=0}^{N} Re(\underline{u}_m \cdot \underline{i}_m^*)$	Active power in kW, where $ku = \frac{\left(\sqrt[\alpha]{\sum_{m=0}^{N} \lvert\underline{u}_m\rvert^{\alpha}}\right)^2}{u_{tot}^2}$ $u_{tot}(f_k) = \sum_{m=0}^{N} \lvert\underline{u}_m\rvert \; ki = \frac{\left(\sqrt[\alpha]{\sum_{m=0}^{N} \lvert\underline{i}_m\rvert^{\alpha}}\right)^2}{i_{tot}^2}$ $i_{tot}(f_k) = \sum_{m=0}^{N} \lvert\underline{i}_m\rvert$
$S = u \cdot i$	Apparent power in kVA
$Q = \sqrt{S^2 - P^2}$	Reactive power in kvar
$cosphi = \frac{P}{S}$	Power factor

Calculation variables: The following calculation variables are available for lines, transformers, series compensation:

$Losses = Re(\underline{S})$	Losses in kW
$loading = i_{max}/i_{nom} \cdot 100$	Loading in %

The following calculation variable is available for machines (static generator, asynchronous machine and synchronous machine):

$loading = S/S_n \cdot 100$	Loading in %, where S_n is the total apparent nominal power in MVA.

Monitor Variables Considering All Frequencies: The following variables are calculated considering all frequencies, f_k:

Intermediate variables:

$i(f_k) = \sqrt[\alpha]{\sum_{m=0}^{N} \lvert\underline{i}_m(f_k)\rvert^{\alpha}}$	Current magnitude in p.u., according to IEC-61000-3-6
$u(f_k) = \sqrt[\alpha]{\sum_{m=0}^{N} \lvert\underline{u}_m(f_k)\rvert^{\alpha}}$	Voltage magnitude in p.u., according to IEC-61000-3-6
$P(f_k) = ku(f_k) \cdot ki(f_k) \cdot \sum_{m=0}^{N} Re(\underline{u}_m(f_k) \cdot \underline{i}_m(f_k)^*)$	Active power in kW, where $ku(f_k) = \frac{\left(\sqrt[\alpha]{\sum_{m=0}^{N} \lvert\underline{u}_m(f_k)\rvert^{\alpha}}\right)^2}{u_{tot}(f_k)^2}$ $u_{tot}(f_k) = \sum_{m=0}^{N} \lvert\underline{u}_m(f_k)\rvert$ $ki(f_k) = \frac{\left(\sqrt[\alpha]{\sum_{m=0}^{N} \lvert\underline{i}_m(f_k)\rvert^{\alpha}}\right)^2}{i_{tot}(f_k)^2}$ $i_{tot}(f_k) = \sum_{m=0}^{N} \lvert\underline{i}_m(f_k)\rvert$

Monitor variables:

$HD = \frac{i(f_{out})}{i(f_1)} \cdot 100$	Harmonic distortion, current in %; based on fundamental frequency values
$HD = \frac{i(f_{out})}{i_{nom}} \cdot 100$	Harmonic distortion, current in %; based on nominal current
$HF = \frac{i(f_{out})}{irms} \cdot 100$	Harmonic factor, current in %
$Irms = \sqrt{\sum_{k\geq 1} i(f_k)^2} \cdot I_{nom}$	Current, RMS in kA
$irms = \frac{Irms}{I_{nom}}$	Current, RMS in p.u.
$THD = \frac{1}{i(f_1)} \cdot \sqrt{irms^2 - i(f_1)^2} \cdot 100$	Total harmonic distortion, current in %; based on fundamental frequency values
$THF = \frac{1}{irms} \cdot \sqrt{irms^2 - i(f_1)^2} \cdot 100$	Total harmonic factor, current in %
$THD = \frac{1}{i_{nom}} \cdot \sqrt{irms^2 - i(f_1)^2} \cdot 100$	Total harmonic distortion, current in %; based on nominal current
$TP = \sum_{k\geq 1} P(f_k)$	Total active power in MW
$TS = \sqrt{Irms^2 \cdot Urms^2 \cdot 3}$	Total apparent power in MVA, where $Urms = \sqrt{\sum_{k\geq 1} u(f_k)^2} \cdot U_{nom}$
$TQ = \sqrt{TS^2 - TP^2}$	Total reactive power in Mvar
$Tcosphi = \frac{TP}{TS}$	Total power factor
$IT = \sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot i(f_k)^2} \cdot I_{nom}$	IT-product in kA
$TAD = \frac{1}{i(f_1)} \cdot \left[\sum_{k\geq 1} i(f_k) - i(f_1)\right] \cdot 100$	Arithmetic distortion in %

Calculation variables: The following calculation variables are available for lines, transformers and series compensation:

$LossesTot = \sum_{h\geq 1}^{H} Losses$	Total losses, including the fundamental in kW, where h is the harmonic order
$LossesHrm = \sum_{h>1}^{H} Losses$	Harmonic losses in kW
$loadingTot = irms_{max}/i_{nom} \cdot 100$	Total loading in %

3.1.0.2 Bus Monitor Variables

Table 3.4 provides the basic nomenclature used throughout this section.

Table 3.4: Nomenclature

Name	Unit	Description
f_k	Hz	Frequency of harmonic order k
f_{out}	Hz	Output frequency specified by user in Harmonic Load Flow command (parameter name: *ifshow*)
$\underline{u}(f_k)$	p.u.	Complex harmonic voltage at frequency of harmonic order k
$u(f_k)$	p.u.	Voltage magnitude at frequency of harmonic order k
U_{nom}	kV	Nominal voltage of the busbar
$\phi_{initial}$	deg	Initial voltage angle
$\underline{Z}$	Ω	Short-circuit impedance (positive or negative sequence, depending on the sequence characteristic of the harmonic order). In the balanced Frequency Sweep, it is the positive sequence impedance.

PowerFactory calculates monitor variables for the Harmonic Analysis command (balanced calculation) in the following manner provided that at least one harmonic source in the network is defined as 'IEC' (TypHmccur parameter name: *i_usym*).

Per IEC source (subscript $m \geq 1$) and per frequency (f_k), the relevant quantities shown in Table 3.5 are available for the calculation of bus monitor variables:

Table 3.5: IEC sources: given quantities

Complex voltage
$\underline{u}_m(f_k)$

For non-IEC sources (subscript $m = 0$), the quantities shown in Table 3.6 are available as a lumped representation of all non-IEC sources, for the calculation of bus monitor variables:

Table 3.6: Non-IEC sources: given quantities

Complex voltage
$\underline{u}_0(f_k)$

Monitor Variables for Output Frequency The following monitor variables are calculated for user-selected output frequency, f_{out}:

Monitor variables:

$R = Re(\underline{Z})$	Resistance in Ω
$X = Im(\underline{Z})$	Network reactance in Ω
$Z = \lvert\underline{Z}\rvert$	Network impedance magnitude in Ω

$phiz = \phi_{\underline{Z}}$	Network impedance angle in deg
$u_= \sqrt[\alpha]{\sum_{m=0}^{N} \lvert \underline{u}_m \rvert^{\alpha}}$	Voltage magnitude in p.u., according to IEC-61000-3-6
$upc = u \cdot 100$	Voltage magnitude in %
$u1 = u$	Voltage magnitude in p.u.
$u1 = 0$	Voltage magnitude in p.u., for harmonic orders 5, 11, 17, 23,. . . ,
$u1pc = u1 \cdot 100$	Voltage magnitude in %
$U = u \cdot U_{nom}/\sqrt{3}$	Line-ground voltage magnitude in kV
$Ul = u \cdot U_{nom}$	Line-line voltage magnitude in kV
$phiu$	(N/A) Voltage angle in deg
$phiurel$	(N/A) Voltage, relative angle in deg
$du = upc - 100$	Voltage deviation in %
$u2 = u$	Negative sequence voltage in p.u., for orders 5, 11, 17, 23,. . . ,
$u2 = 0$	Negative sequence voltage in p.u.

Monitor Variables Considering All Frequencies: The following variables are calculated considering all frequencies, f_k:

Intermediate variables:

$u(f_k)_= \sqrt[\alpha]{\sum_{m=0}^{N} \lvert \underline{u}_m(f_k) \rvert^{\alpha}}$	Voltage magnitude in p.u., according to IEC-61000-3-6

Monitor variables:

$urms = \sqrt{\sum_{k \geq 1} u(f_k)^2}$	RMS value of voltage in p.u.
$HD = \frac{u(f_{out})}{u(f_1)} \cdot 100$	Harmonic distortion in %; based on fundamental frequency values
$HD = \frac{u(f_{out})}{u_{nom}} \cdot 100$	Harmonic distortion in %; based on nominal voltage, where $u_{nom} = 1$p.u.
$HF = \frac{u(f_{out})}{urms} \cdot 100$	Harmonic factor in %

$THD = \frac{1}{u(f_1)} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic distortion in %; based on fundamental frequency values
$THD = \frac{1}{u_{nom}} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic distortion in %; based on nominal voltage, where $u_{nom} = 1$p.u.
$THF = \frac{1}{urms} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic factor in %
$THF = \frac{1}{urms} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic factor in %
$Ulrms = urms \cdot U_{nom}$	RMS value of line-line voltage in kV
$Urms = urms \cdot U_{nom}/\sqrt{3}$	RMS value of line-ground voltage in kV
$TAD = \frac{1}{u(f_1)} \cdot \left[\sum_{k \geq 1} u(f_k) - u(f_1)\right] \cdot 100$	Arithmetic distortion in %
$THF = \frac{1}{urms} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic factor in %
$uasum = \sum_{k=1}^{n} u(f_k)$	Arithmetic voltage sum in p.u.
$TIF = \frac{1}{urms} \sqrt{\sum_{k=1}^{n} tif(f_k)^2 \cdot u(f_k)^2}$	Telephone interference factor
$urmsint = \sqrt{\sum_{\substack{k \geq 2 \\ integer}} u(f_k)^2}$	RMS value of integer harmonics in p.u.
$uasumint = \sum_{\substack{k \geq 2 \\ integer}} u(f_k)$	Arithmetic sum of integer harmonics in p.u.
$THDint = \frac{1}{u(f_1)} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic distortion in %, integer orders, (Fundamental frequency values)
$THDint = \frac{1}{u_{nom}} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic distortion in %, integer orders (nominal voltage), where $u_{nom} = 1$p.u. and $urms = \sqrt{\sum_{\substack{k \geq 1 \\ integer}} u(f_k)^2}$
$THDnint = \frac{1}{u(f_1)} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic distortion in %, non-integer orders, (Fundamental frequency values)
$THDnint = \frac{1}{u_{nom}} \sqrt{urms^2 - u(f_1)^2} \cdot 100$	Total harmonic distortion in %, non-integer orders (nominal voltage), where $u_{nom} = 1$p.u. and $urms = \sqrt{\sum_{\substack{k \geq 1 \\ non-integer}} u(f_k)^2}$

$TADint = \frac{1}{u(f_1)} \cdot \left[\sum_{\substack{k \geq 1 \\ integer}} u(f_k) - u(f_1) \right] \cdot 100$	Arithmetic distortion in %, integer orders
$TADnint = \frac{1}{u(f_1)} \cdot \left[\sum_{\substack{k \geq 1 \\ non-integer}} u(f_k) - u(f_1) \right] \cdot 100$	Arithmetic distortion in %, non-integer orders

3.2 Unbalanced Calculation

3.2.0.1 Branch Monitor Variables

Table 3.7 provides the basic nomenclature used throughout this section.

Table 3.7: Nomenclature

Name	Unit	Description
f_k		Frequency at harmonic order k
f_{out}		Output frequency specified by user in Harmonic Load Flow Command (parameter name: *ifshow*)
$\underline{i}_a(f_k)$	p.u.	Complex harmonic current for phase A, at frequency of harmonic order k (all p.u. values are based on I_{nom}). Similar nomenclature follows for other phases.
$\underline{u}_a(f_k)$	p.u.	Complex harmonic voltage for phase B, at frequency of harmonic order k (all p.u. values are based on U_{nom}). Similar nomenclature follows for other phases.
$\underline{I}_a(f_k)$	A	Complex harmonic current for phase A, at frequency of harmonic order k. Similar nomenclature follows for other phases.
$\underline{U}_a(f_k)$	kV	Complex harmonic voltage for phase A, at frequency of harmonic order k. Similar nomenclature follows for other phases.
$\underline{i}_0(f_k)$	p.u.	Complex zero sequence current, at frequency of harmonic order k.
$\underline{u}_0(f_k)$	p.u.	Complex zero sequence voltage, at frequency of harmonic order k.
$i{:}A(f_k)$	p.u.	Current magnitude at frequency of harmonic order k. Similar nomenclature follows for other phases.
$i0(f_k)$	p.u.	Zero sequence current magnitude at frequency of harmonic order k. Similar nomenclature follows for positive and negative sequence currents.
I_{nom}, i_{nom}	A, p.u.	Nominal current of element
U_{nom}, u_{nom}	kV, p.u.	Rated voltage of element
$tif(f_k)$	-	Telephone influence factor (according to IEEE) at frequency of harmonic order k.

PowerFactory calculates monitor variables for the Harmonic Analysis command (unbalanced calculation) in the following manner provided that at least one harmonic source in the network is defined as 'IEC' (TypHmccur parameter name: *i_usym*).

Per IEC source (subscript $m \geq 1$) and per frequency (f_k), the quantities shown in Table 3.8 are

available for the calculation of branch monitor variables:

Table 3.8: IEC sources: given quantities

Complex phase voltages	Complex phase currents
$\underline{u}_{a,m}(f_k)$	$\underline{i}_{a,m}(f_k)$
$\underline{u}_{b,m}(f_k)$	$\underline{i}_{b,m}(f_k)$
$\underline{u}_{c,m}(f_k)$	$\underline{i}_{c,m}(f_k)$
$\underline{u}_{n,m}(f_k)$	$\underline{i}_{n,m}(f_k)$

For non-IEC sources (subscript $m = 0$), the quantities shown in Table 3.9 are available as a lumped representation of all non-IEC sources, for the calculation of branch monitor variables:

Table 3.9: Non-IEC sources: given quantities

Complex phase voltage	Complex phase current
$\underline{u}_{a,0}(f_k)$	$\underline{i}_{a,0}(f_k)$
$\underline{u}_{b,0}(f_k)$	$\underline{i}_{b,0}(f_k)$
$\underline{u}_{c,0}(f_k)$	$\underline{i}_{c,0}(f_k)$
$\underline{u}_{n,0}(f_k)$	$\underline{i}_{n,0}(f_k)$

Monitor Variables for Output Frequency The following monitor variables are calculated for user-selected output frequency, f_{out}:

Intermediate variables:

$u{:}A = \sqrt[\alpha]{\sum_{m=0}^{N} \left|\underline{u}_{a,m}\right|^{\alpha}}$ Phase voltage magnitudes in p.u. according to IEC-61000-3-6

$u{:}B = \sqrt[\alpha]{\sum_{m=0}^{N} \left|\underline{u}_{b,m}\right|^{\alpha}}$

$u{:}C = \sqrt[\alpha]{\sum_{m=0}^{N} \left|\underline{u}_{c,m}\right|^{\alpha}}$

$u{:}N = \sqrt[\alpha]{\sum_{m=0}^{N} \left|\underline{u}_{n,m}\right|^{\alpha}}$

Monitor variables:

$i{:}A_{=} \sqrt[\alpha]{\sum_{m=0}^{N} \left|\underline{i}_{a,m}\right|^{\alpha}}$ Phase current magnitudes in p.u. according to IEC-61000-3-6

$i{:}B_{=} \sqrt[\alpha]{\sum_{m=0}^{N} \left|\underline{i}_{b,m}\right|^{\alpha}}$

$i{:}C_{=} \sqrt[\alpha]{\sum_{m=0}^{N} \left|\underline{i}_{c,m}\right|^{\alpha}}$

$i{:}N_{=} \sqrt[\alpha]{\sum_{m=0}^{N} \left|\underline{i}_{n,m}\right|^{\alpha}}$

$I{:}A = \|i{:}A\| \cdot I_{nom}$	Phase currents in A
$I{:}B = \|i{:}B\| \cdot I_{nom}$	
$I{:}C = \|i{:}C\| \cdot I_{nom}$	
$I{:}N = \|i{:}N\| \cdot I_{nom}$	
$phiui{:}A$	(N/A) Angle between voltage and current in deg
$phiui{:}B$	(N/A)
$phiui{:}C$	(N/A)
$phiui{:}N$	(N/A)
$phii{:}A$	(N/A) Current angle, absolute in deg
$phii{:}B$	(N/A)
$phii{:}C$	(N/A)
$phii{:}N$	(N/A)
$P{:}A = \sum_{m=0}^{N} Re(\underline{u}_{a,m} \cdot \underline{i}^*_{a,m})$	Active power in kW
$P{:}B = \sum_{m=0}^{N} Re(\underline{u}_{b,m} \cdot \underline{i}^*_{b,m})$	
$P{:}C = \sum_{m=0}^{N} Re(\underline{u}_{c,m} \cdot \underline{i}^*_{c,m})$	
$P{:}N = \sum_{m=0}^{N} Re(\underline{u}_{n,m} \cdot \underline{i}^*_{n,m})$	
$S{:}A = u{:}A \cdot i{:}A$	Apparent power in kVA
$S{:}B = u{:}B \cdot i{:}B$	
$S{:}C = u{:}C \cdot i{:}C$	
$S{:}N = u{:}N \cdot i{:}N$	
$Q{:}A = \sqrt{S{:}A^2 - P{:}A^2}$	Reactive power in kvar
$Q{:}B = \sqrt{S{:}B^2 - P{:}B^2}$	
$Q{:}C = \sqrt{S{:}C^2 - P{:}C^2}$	
$Q{:}N = \sqrt{S{:}N^2 - P{:}N^2}$	

$cosphi{:}A$	(N/A) Power factor
$cosphi{:}B$	(N/A)
$cosphi{:}C$	(N/A)
$cosphi{:}N$	(N/A)
$i0 = \sqrt[\alpha]{\sum_{m=0}^{N} \left\lvert \underline{i}_{0,m} \right\rvert^{\alpha}}$	Zero sequence current in p.u., where $\underline{i}_{0,m} = \frac{1}{3}(\underline{i}_{a,m} + \underline{i}_{b,m} + \underline{i}_{c,m})$
$I0 = i0 \cdot I_{nom}$	Zero sequence current in A
$phiu0i0$	(N/A) Angle between voltage and current in zero sequence system in deg
$i1 = \sqrt[\alpha]{\sum_{m=0}^{N} \left\lvert \underline{i}_{1,m} \right\rvert^{\alpha}}$	Positive sequence current in p.u., where $\underline{i}_{1,m} = \frac{1}{3}(\underline{i}_{a,m} + a \cdot \underline{i}_{b,m} + a^2 \cdot \underline{i}_{c,m})$ and $a = \angle 120°$
$I1 = i1 \cdot I_{nom}$	Positive sequence current in A
$phiu1i1$	(N/A) Angle between voltage and current in positive sequence system in deg
$i2 = \sqrt[\alpha]{\sum_{m=0}^{N} \left\lvert \underline{i}_{2,m} \right\rvert^{\alpha}}$	Negative sequence current in p.u. where $\underline{i}_{2,m} = \frac{1}{3}(\underline{i}_{a,m} + a^2 \cdot \underline{i}_{b,m} + a \cdot \underline{i}_{c,m})$ and $a = \angle 120°$
$I2 = i2 \cdot I_{nom}$	Negative sequence current in A
$phiu2i2$	(N/A) Angle between voltage and current in negative sequence system in deg
$Psum = P{:}A + P{:}B + P{:}C + P{:}N$	Active power in kW
$Qsum = Q{:}A + Q{:}B + Q{:}C + Q{:}N$	Reactive power in kvar
$Ssum = \left\lvert \sum_{x=[A,B,C,N]} \sqrt{P{:}x^2 + Q{:}x^2} \right\rvert$	Apparent power in kVA
$cosphisum = \frac{Psum}{Ssum}$	Power factor
$ubfac = \frac{i2}{i1}$	Unbalance factor

Calculation variables: The following calculation variables are available for lines:

$LossesPh{:}A = Re(\underline{S}_a)$	Losses in kW
$LossesPh{:}B = Re(\underline{S}_b)$	
$LossesPh{:}C = Re(\underline{S}_c)$	
$LossesPh{:}N = Re(\underline{S}_n)$	
$Losses = \sum_{x=[A,B,C,N]} LossesPh{:}x$	Losses in kW
$loadingPh{:}A = i{:}A_{max}/i_{nom} \cdot 100$	Loading in %
$loadingPh{:}B = i{:}B_{max}/i_{nom} \cdot 100$	
$loadingPh{:}C = i{:}C_{max}/i_{nom} \cdot 100$	
$loadingPh{:}N = i{:}N_{max}/i_{nom} \cdot 100$	
$loading = i_{max}/i_{nom} \cdot 100$	Loading in %

The following calculation variables are available for transformers and series compensation:

$Losses = Re(\underline{S})$	Losses in kW
$loading = i_{max}/i_{nom} \cdot 100$	Loading in %

The following calculation variable is available for machines (static generator, asynchronous machine and synchronous machine):

$loading = S/S_n \cdot 100$	Loading in %, where S_n is the total apparent nominal power in MVA.

Monitor Variables Considering All Frequencies: The following monitor variables are calculated considering all frequencies f_k:

Intermediate variables:

$i{:}A(f_k) = \sqrt[\alpha]{\sum_{m=0}^{N} \left\lvert \underline{i}_{a,m}(f_k) \right\rvert^{\alpha}}$	Phase current magnitudes in p.u. according to IEC-61000-3-6
$i{:}B(f_k) = \sqrt{[\alpha] \sum_{m=0}^{N} \left\lvert \underline{i}_{b,m}(f_k) \right\rvert^{\alpha}}$	
$i{:}C(f_k) = \sqrt{[\alpha] \sum_{m=0}^{N} \left\lvert \underline{i}_{c,m}(f_k) \right\rvert^{\alpha}}$	
$i{:}N(f_k) = \sqrt{[\alpha] \sum_{m=0}^{N} \left\lvert \underline{i}_{n,m}(f_k) \right\rvert^{\alpha}}$	

$u{:}A(f_k) = \sqrt{[\alpha] \sum_{m=0}^{N} \left|\underline{u}_{a,m}(f_k)\right|^{\alpha}}$ Phase voltage magnitudes in p.u. according to IEC-61000-3-6

$u{:}B(f_k) = \sqrt{[\alpha] \sum_{m=0}^{N} \left|\underline{u}_{b,m}(f_k)\right|^{\alpha}}$

$u{:}C(f_k) = \sqrt{[\alpha] \sum_{m=0}^{N} \left|\underline{u}_{c,m}(f_k)\right|^{\alpha}}$

$u{:}N(f_k) = \sqrt{[\alpha] \sum_{m=0}^{N} \left|\underline{u}_{n,m}(f_k)\right|^{\alpha}}$

$i0(f_k) = \sqrt{[\alpha] \sum_{m=0}^{N} \left|\underline{i}_{0,m}(f_k)\right|^{\alpha}}$ Zero sequence current in p.u. where $\underline{i}_{0,m}(f_k) = \frac{1}{3}(\underline{i}_{a,m}(f_k) + \underline{i}_{b,m}(f_k) + \underline{i}_{c,m}(f_k))$ and $a = \angle 120^\circ$

$i1(f_k) = \sqrt{[\alpha] \sum_{m=0}^{N} \left|\underline{i}_{1,m}(f_k)\right|^{\alpha}}$ Positive sequence current in p.u. where $\underline{i}_{1,m}(f_k) = \frac{1}{3}(\underline{i}_{a,m}(f_k) + a \cdot \underline{i}_{b,m}(f_k) + a^2 \cdot \underline{i}_{c,m}(f_k))$ and $a = \angle 120^\circ$

$i2(f_k) = \sqrt{[\alpha] \sum_{m=0}^{N} \left|\underline{i}_{2,m}(f_k)\right|^{\alpha}}$ Negative sequence current in p.u. where $\underline{i}_{2,m}(f_k) = \frac{1}{3}(\underline{i}_{a,m}(f_k) + a^2 \cdot \underline{i}_{b,m}(f_k) + a \cdot \underline{i}_{c,m}(f_k))$ and $a = \angle 120^\circ$

$P{:}A(f_k) = \sum_{m=0}^{N} Re(\underline{u}_{a,m}(f_k) \cdot \underline{i}_{a,m}(f_k)^*)$ Active power in kW

$P{:}B(f_k) = \sum_{m=0}^{N} Re(\underline{u}_{b,m}(f_k) \cdot \underline{i}_{b,m}(f_k)^*)$

$P{:}C(f_k) = \sum_{m=0}^{N} Re(\underline{u}_{c,m}(f_k) \cdot \underline{i}_{c,m}(f_k)^*)$

$P{:}N(f_k) = \sum_{m=0}^{N} Re(\underline{u}_{n,m}(f_k) \cdot \underline{i}_{n,m}(f_k)^*)$

Monitor variables:

$irms{:}A = \sqrt{\sum_{k \geq 1} i{:}A(f_k)^2}$ Phase currents in p.u. based on rated current of element

$irms{:}B = \sqrt{\sum_{k \geq 1} i{:}B(f_k)^2}$

$irms{:}C = \sqrt{\sum_{k \geq 1} i{:}C(f_k)^2}$

$irms{:}N = \sqrt{\sum_{k \geq 1} i{:}N(f_k)^2}$

$Irms{:}A = irms{:}A \cdot I_{nom}$	Phase currents in A
$Irms{:}B = irms{:}B \cdot I_{nom}$	
$Irms{:}C = irms{:}C \cdot I_{nom}$	
$Irms{:}N = irms{:}N \cdot I_{nom}$	
$HD{:}A = \frac{i{:}A(f_{out})}{i{:}A(f_1)} \cdot 100$	Harmonic distortion in %; based on fundamental frequency values
$HD{:}B = \frac{i{:}B(f_{out})}{i{:}B(f_1)} \cdot 100$	
$HD{:}C = \frac{i{:}C(f_{out})}{i{:}C(f_1)} \cdot 100$	
$HD{:}N = \frac{i{:}N(f_{out})}{i{:}N(f_1)} \cdot 100$	
$HD{:}A = \frac{i{:}A(f_{out})}{i_{nom}(f_1)} \cdot 100$	Harmonic distortion in %; based on nominal current
$HD{:}B = \frac{i{:}B(f_{out})}{i_{nom}(f_1)} \cdot 100$	
$HD{:}C = \frac{i{:}C(f_{out})}{i_{nom}(f_1)} \cdot 100$	
$HD{:}N = \frac{i{:}N(f_{out})}{i_{nom}(f_1)} \cdot 100$	
$HF{:}A = \frac{i{:}A(f_{out})}{irms{:}A(f_1)} \cdot 100$	Harmonic factor in %
$HF{:}B = \frac{i{:}B(f_{out})}{irms{:}B(f_1)} \cdot 100$	
$HF{:}C = \frac{i{:}C(f_{out})}{irms{:}C(f_1)} \cdot 100$	
$HF{:}N = \frac{i{:}N(f_{out})}{irms{:}N(f_1)} \cdot 100$	
$THD{:}A = \frac{1}{i{:}A(f_1)} \cdot \sqrt{irms{:}A^2 - i{:}A(f_1)^2} \cdot 100$	Total harmonic distortion, current in %; based on fundamental frequency values
$THD{:}B = \frac{1}{i{:}B(f_1)} \cdot \sqrt{irms{:}B^2 - i{:}B(f_1)^2} \cdot 100$	
$THD{:}C = \frac{1}{i{:}C(f_1)} \cdot \sqrt{irms{:}C^2 - i{:}C(f_1)^2} \cdot 100$	
$THD{:}N = \frac{1}{i{:}N(f_1)} \cdot \sqrt{irms{:}N^2 - i{:}N(f_1)^2} \cdot 100$	

$THD{:}A = \frac{1}{i_{nom}} \cdot \sqrt{irms{:}A^2 - i{:}A(f_1)^2} \cdot 100$ — Total harmonic distortion, current in %; based on nominal current

$THD{:}B = \frac{1}{i_{nom}} \cdot \sqrt{irms{:}B^2 - i{:}B(f_1)^2} \cdot 100$

$THD{:}C = \frac{1}{i_{nom}} \cdot \sqrt{irms{:}C^2 - i{:}C(f_1)^2} \cdot 100$

$THD{:}N = \frac{1}{i_{nom}} \cdot \sqrt{irms{:}N^2 - i{:}N(f_1)^2} \cdot 100$

$THF{:}A = \frac{1}{irms{:}A} \cdot \sqrt{irms{:}A^2 - i{:}A(f_1)^2} \cdot 100$ — Total harmonic factor, current in %

$THF{:}B = \frac{1}{irms{:}B} \cdot \sqrt{irms{:}B^2 - i{:}B(f_1)^2} \cdot 100$

$THF{:}C = \frac{1}{irms{:}C} \cdot \sqrt{irms{:}C^2 - i{:}C(f_1)^2} \cdot 100$

$THF{:}N = \frac{1}{irms{:}N} \cdot \sqrt{irms{:}N^2 - i{:}N(f_1)^2} \cdot 100$

$TP{:}A = \sum_{k \geq 1} P{:}A(f_k)$ — Total active power in MW

$TP{:}B = \sum_{k \geq 1} P{:}B(f_k)$

$TP{:}C = \sum_{k \geq 1} P{:}C(f_k)$

$TP{:}N = \sum_{k \geq 1} P{:}N(f_k)$

$TS{:}A = \sqrt{Irms{:}A^2 \cdot Urms{:}A^2}$ — Total apparent power in MVA, where $Urms{:}A = \sqrt{\sum_{k \geq 1} u{:}A(f_k)^2} \cdot U_{nom}$

$TS{:}B = \sqrt{Irms{:}B^2 \cdot Urms{:}B^2}$

$TS{:}C = \sqrt{Irms{:}C^2 \cdot Urms{:}C^2}$

$TS{:}N = \sqrt{Irms{:}N^2 \cdot Urms{:}N^2}$

$TQ{:}A = \sqrt{TS{:}A^2 - TP{:}A^2}$ — Total reactive power in Mvar

$TQ{:}B = \sqrt{TS{:}B^2 - TP{:}B^2}$

$TQ{:}C = \sqrt{TS{:}C^2 - TP{:}C^2}$

$TQ{:}N = \sqrt{TS{:}N^2 - TP{:}N^2}$

$Tcosphi{:}A = \frac{TP:A}{TS:A}$	Total power factor
$Tcosphi{:}B = \frac{TP:B}{TS:B}$	
$Tcosphi{:}C = \frac{TP:C}{TS:C}$	
$Tcosphi{:}N = \frac{TP:N}{TS:N}$	
$TAD{:}A = \frac{1}{i{:}A(f_1)} \cdot \left[\sum_{k\geq 1} i{:}A(f_k) - i{:}A(f_1)\right] \cdot 100$	Arithmetic distortion in %
$TAD{:}B = \frac{1}{i{:}B(f_1)} \cdot \left[\sum_{k\geq 1} i{:}B(f_k) - i{:}B(f_1)\right] \cdot 100$	
$TAD{:}C = \frac{1}{i{:}C(f_1)} \cdot \left[\sum_{k\geq 1} i{:}C(f_k) - i{:}C(f_1)\right] \cdot 100$	
$TAD{:}N = \frac{1}{i{:}N(f_1)} \cdot \left[\sum_{k\geq 1} i{:}N(f_k) - i{:}N(f_1)\right] \cdot 100$	
$IT{:}A = \sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot i{:}A(f_k)^2} \cdot I_{nom}$	IT-product in kA
$IT{:}B = \sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot i{:}B(f_k)^2} \cdot I_{nom}$	
$IT{:}C = \sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot i{:}C(f_k)^2} \cdot I_{nom}$	
$IT{:}N = \sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot i{:}N(f_k)^2} \cdot I_{nom}$	
$I0rms = \sqrt{\sum_{k\geq 1} i0(f_k)^2} \cdot I_{nom}$	Zero sequence current, RMS value in kA
$i0rms = \frac{I0rms}{I_{nom}}$	Zero sequence current, RMS value in p.u.
$THD0 = \frac{1}{i0(f_1)} \sqrt{i0rms^2 - i0(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, zero sequence); based on fundamental frequency values
$THF0 = \frac{1}{i0rms} \sqrt{i0rms^2 - i0(f_1)^2} \cdot 100$	Total harmonic factor in % (current, zero sequence)
$THD0 = \frac{1}{i_{nom}} \sqrt{i0rms^2 - i0(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, zero sequence); based on nominal current
$I1rms = \sqrt{\sum_{k\geq 1} i1(f_k)^2} \cdot I_{nom}$	Positive sequence current, RMS value in kA
$i1rms = \frac{I1rms}{I_{nom}}$	Positive sequence current, RMS value in p.u.
$THD1 = \frac{1}{i1(f_1)} \sqrt{i1rms^2 - i1(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, positive sequence); based on fundamental frequency values
$THD1 = \frac{1}{i_{nom}} \sqrt{i1rms^2 - i1(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, positive sequence); based on nominal current

$THF1 = \frac{1}{i1rms}\sqrt{i1rms^2 - i1(f_1)^2} \cdot 100$	Total harmonic factor in % (current, positive sequence)
$I2rms = \sqrt{\sum_{k\geq 1} i2(f_k)^2} \cdot I_{nom}$	Negative sequence current, RMS value in kA
$i2rms = \frac{I2rms}{I_{nom}}$	Negative sequence current, RMS value in p.u.
$THD2 = \frac{1}{i2(f_1)}\sqrt{i2rms^2 - i2(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, negative sequence); based on fundamental frequency values
$THD2 = \frac{1}{i_{nom}}\sqrt{i2rms^2 - i2(f_1)^2} \cdot 100$	Total harmonic distortion in % (current, negative sequence)l; based on nominal current
$THF2 = \frac{1}{i2rms}\sqrt{i2rms^2 - i2(f_1)^2} \cdot 100$	Total harmonic factor in % (current, negative sequence)
$HD0 = \frac{i0(f_{out})}{\sqrt{i0(f_1)^2+i1(f_1)^2+i2(f_1)^2}} \cdot 100$	Harmonic distortion, zero sequence in %; based on fundamental frequency values
$HD1 = \frac{i1(f_{out})}{\sqrt{i0(f_1)^2+i1(f_1)^2+i2(f_1)^2}} \cdot 100$	Harmonic distortion, positive sequence in %
$HD2 = \frac{i2(f_{out})}{\sqrt{i0(f_1)^2+i1(f_1)^2+i2(f_1)^2}} \cdot 100$	Harmonic distortion, negative sequence in %
$HD0 = \frac{i0(f_{out})}{i_{nom}} \cdot 100$	Harmonic distortion, zero sequence in %; based on nominal current
$HD1 = \frac{i1(f_{out})}{i_{nom}} \cdot 100$	Harmonic distortion, positive sequence in %
$HD2 = \frac{i2(f_{out})}{i_{nom}} \cdot 100$	Harmonic distortion, negative sequence in %
$HF0 = \frac{i0(f_{out})}{i0rms} \cdot 100$	Harmonic factor, zero sequence in %
$HF1 = \frac{i1(f_{out})}{i1rms} \cdot 100$	Harmonic factor, positive sequence in %
$HF2 = \frac{i2(f_{out})}{i2rms} \cdot 100$	Harmonic factor, negative sequence in %
$TPsum = \sum_{x=[A,B,C,N]} TP{:}x$	Total active power in MW
$TQsum = \sum_{x=[A,B,C,N]} TQ{:}x$	Total reactive power in Mvar
$TSsum = \sqrt{TPsum^2 + TQsum^2}$	Total apparent power in MVA
$Tcosphisum = \frac{TPsum}{TSsum}$	Total power factor

$THDbal = \sqrt{THD1^2 + THD2^2}$	Total harmonic distortion in %, excluding zero sequence
$THDtot = \sqrt{THD1^2 + THD2^2 + THD0^2}$	Total harmonic distortion in %, including zero sequence
$TADmx = \max_{x=[A,B,C,N]}(TAD{:}x)$	Arithmetic distortion in %
$TADint = \max_{x=[A,B,C,N]}(TADint{:}x)$	Arithmetic distortion in %, where $TADint{:}x = \frac{1}{i{:}x(f_1)} \cdot \left[\sum_{\substack{k\geq 1 \\ integer}} i{:}x(f_k) - i{:}x(f_1) \right] \cdot 100$
$TADnint = \max_{x=[A,B,C,N]}(TADnint{:}x)$	Arithmetic distortion in %, where $TADnint{:}x = \frac{1}{i{:}x(f_1)} \cdot \left[\sum_{\substack{k>1 \\ non-integer}} i{:}x(f_k) - i{:}x(f_1) \right] \cdot 100$
$TAD_3 = \max_{x=[A,B,C,N]}(TADint{:}x)$	Arithmetic distortion in %, harmonic orders: multiples of three
$TAD_2 = \max_{x=[A,B,C,N]}(TADint{:}x)$	Arithmetic distortion in %, harmonic orders: multiples of two
$THDint = \max_{x=[A,B,C,N]}(THDint{:}x)$	Total harmonic distortion in %, where $THDint{:}x = \frac{1}{i{:}x(f_1)} \cdot \sqrt{irms{:}x^2 - i{:}x(f_1)^2} \cdot 100$ (Fundamental frequency values) or $THDint{:}x = \frac{1}{inom} \cdot \sqrt{irms{:}x^2 - i{:}x(f_1)^2} \cdot 100$ (Nominal current) and $irms{:}x = \sqrt{\sum_{\substack{k\geq 1 \\ integer}} i{:}x(f_k)^2}$
$THDnint = \max_{x=[A,B,C,N]}(THDnint{:}x)$	Total harmonic distortion in %, where $THDnint{:}x = \frac{1}{i{:}x(f_1)} \cdot \sqrt{irms{:}x^2 - i{:}x(f_1)^2} \cdot 100$ (Fundamental frequency values) or $THDnint{:}x = \frac{1}{inom} \cdot \sqrt{irms{:}x^2 - i{:}x(f_1)^2} \cdot 100$ (Nominal current) and $irms{:}x = \sqrt{\sum_{\substack{k>1 \\ non-integer}} i{:}x(f_k)^2}$

$THD_3 = \max_{x=[A,B,C,N]}(THDint{:}x)$	Total harmonic distortion in %, harmonic orders: multiples of three
$THD_2 = \max_{x=[A,B,C,N]}(TADint{:}x)$	Total harmonic distortion in %, harmonic orders: multiples of two
$ITmx = \max(IT{:}x)$	IT-Product in kA

Calculation variables: The following calculation variables are available for lines, transformers, series compensation:

$LossesTot = \sum_{h\geq 1}^{H} \sum_{x=[A,B,C,N]} Losses{:}x$	Total losses, including the fundamental in kW, where h is the harmonic order
$LossesHrm = \sum_{h>1}^{H} \sum_{x=[A,B,C,N]} Losses{:}x$	Harmonic losses in kW
$loadingTot = irms_{max}/i_{nom} \cdot 100$	Total loading in %

3.2.0.2 Bus Monitor Variables

Table 3.10 provides the nomenclature used in this section.

Table 3.10: Nomenclature

Name	Unit	Description
f_k	Hz	Frequency of harmonic order k
f_{out}	Hz	Output frequency specified by user in Harmonic Load Flow Command (parameter name: *ifshow*)
$\underline{u}_a(f_k)$	p.u.	Complex harmonic voltage, phase A. Similar nomenclature is used for the other phases.
$\underline{ul}_a(f_k)$	p.u.	Complex line-line voltage, phase A, where $\underline{ul}_a(f_k) = \underline{u}_a(f_k) - \underline{u}_b(f_k)$
$ul{:}A(f_k)$	p.u.	Line-line voltage magnitude, phase A, where $ul{:}A(f_k) = \lvert\underline{u}_a(f_k) - \underline{u}_b(f_k)\rvert$. Similar follows for the other phases.
$\underline{uln}_a(f_k)$	p.u.	Complex line-neutral voltage, phase A, where $\underline{uln}_a(f_k) = \underline{u}_a(f_k) - \underline{u}_n(f_k)$. Similar follows for the other phases.
$uln{:}A(f_k)$	p.u.	Line-line voltage magnitude, phase A, where $uln{:}A(f_k) = \lvert\underline{u}_a(f_k) - \underline{u}_n(f_k)\rvert$. Similar follows for the other phases.
$u{:}A(f_k)$	p.u.	Voltage magnitude at frequency of harmonic order k
U_{nom}	kV	Nominal voltage of the busbar
$\underline{Zl}_a$	Ω	Line impedance, phase A. Similar follows for the other phases.
$\underline{Zln}_a$	Ω	Line-neutral impedance, phase A. Similar follows for the other phases.
$\underline{Z}_a$	Ω	Short-circuit impedance, phase A. Similar nomenclature is used for other phases.

PowerFactory calculates monitor variables for the Harmonic Analysis command (unbalanced calculation) in the following manner provided that at least one harmonic source in the network is defined as 'IEC' (TypHmccur parameter name: *i_usym*).

Per IEC source (subscript $m \geq 1$) and per frequency (f_k), the relevant quantities shown in Table 3.11 are available for the calculation of bus monitor variables:

Table 3.11: IEC sources: given quantities

Complex phase voltage
$\underline{u}_{a,m}(f_k)$
$\underline{u}_{b,m}(f_k)$
$\underline{u}_{c,m}(f_k)$

For non-IEC sources (subscript $m = 0$), the quantities shown in Table 3.12 are available as a

lumped representation of all non-IEC sources, for the calculation of bus monitor variables:

Table 3.12: Non-IEC sources: given quantities

Complex phase voltage
$\underline{u}_{a,0}(f_k)$
$\underline{u}_{b,0}(f_k)$
$\underline{u}_{c,0}(f_k)$

Monitor Variables for Output Frequency The following monitor variables are calculated for user-selected output frequency, f_{out}:

Intermediate variables:

$u{:}A = \sqrt{[\alpha] \sum_{m=0}^{N} \left\lvert \underline{u}_{a,m} \right\rvert^{\alpha}}$	Line-ground voltage, magnitude in p.u., according to IEC-61000-3-6
$u{:}B = \sqrt{[\alpha] \sum_{m=0}^{N} \left\lvert \underline{u}_{b,m} \right\rvert^{\alpha}}$	
$u{:}C = \sqrt{[\alpha] \sum_{m=0}^{N} \left\lvert \underline{u}_{c,m} \right\rvert^{\alpha}}$	
$u{:}A = \sqrt{[\alpha] \sum_{m=0}^{N} \left\lvert \underline{u}_{a,m} \right\rvert^{\alpha}}$	Line-ground voltage, magnitude in p.u., according to IEC-61000-3-6
$u{:}B = \sqrt{[\alpha] \sum_{m=0}^{N} \left\lvert \underline{u}_{b,m} \right\rvert^{\alpha}}$	
$u{:}C = \sqrt{[\alpha] \sum_{m=0}^{N} \left\lvert \underline{u}_{c,m} \right\rvert^{\alpha}}$	
$upc{:}A = u{:}A \cdot 100$	Line-ground voltage, magnitude in %
$upc{:}B = u{:}B \cdot 100$	
$upc{:}C = u{:}C \cdot 100$	
$U{:}A = u{:}A \cdot U_{nom}/\sqrt{3}$	Line-ground voltage magnitude in kV
$U{:}B = u{:}B \cdot U_{nom}/\sqrt{3}$	
$U{:}C = u{:}C \cdot U_{nom}/\sqrt{3}$	

Monitor variables:

$phiu$:A	(N/A) Voltage angle in deg
$phiu$:B	(N/A)
$phiu$:C	(N/A)
$phiurel$:A	(N/A) Voltage, relative angle in deg
$phiurel$:B	(N/A)
$phiurel$:C	(N/A)
ul:$A = \sqrt{[\alpha] \sum_{m=0}^{N} \left\|\underline{ul}_{a,m}\right\|^{\alpha}}$	Line-line voltage magnitude in p.u.
ul:$B = \sqrt{[\alpha] \sum_{m=0}^{N} \left\|\underline{ul}_{b,m}\right\|^{\alpha}}$	
ul:$C = \sqrt{[\alpha] \sum_{m=0}^{N} \left\|\underline{ul}_{c,m}\right\|^{\alpha}}$	
$ulpc$:$A = u$:$A \cdot 100$	Line-line voltage magnitude in %
$ulpc$:$B = u$:$B \cdot 100$	
$ulpc$:$C = u$:$C \cdot 100$	
Ul:$A = u$:$A \cdot U_{nom}$	Line-line voltage magnitude in kV
Ul:$B = u$:$B \cdot U_{nom}$	
Ul:$C = u$:$C \cdot U_{nom}$	
$phiul$:A	(N/A) Line-line voltage angle in deg
$phiul$:B	(N/A)
$phiul$:C	(N/A)
uln:$A = \sqrt{[\alpha] \sum_{m=0}^{N} \left\|\underline{uln}_{a,m}\right\|^{\alpha}}$	Line-neutral voltage magnitude in p.u.
uln:$B = \sqrt{[\alpha] \sum_{m=0}^{N} \left\|\underline{uln}_{b,m}\right\|^{\alpha}}$	
uln:$C = \sqrt{[\alpha] \sum_{m=0}^{N} \left\|\underline{uln}_{c,m}\right\|^{\alpha}}$	

$Uln{:}A = uln{:}A \cdot Unom/\sqrt{3}$	Line-neutral voltage magnitude in kV
$Uln{:}B = uln{:}B \cdot Unom/\sqrt{3}$	
$Uln{:}C = uln{:}C \cdot Unom/\sqrt{3}$	
$phiuln{:}A$	(N/A) Line-neutral voltage angle in deg
$phiuln{:}B$	(N/A)
$phiuln{:}C$	(N/A)
$Upht{:}A = uln{:}A \cdot Unom$	Phase technology dependent voltage magnitude in kV
$Upht{:}B = uln{:}B \cdot Unom$	
$Upht{:}C = uln{:}C \cdot Unom$	
$R{:}A = \Re(\underline{Z}_a)$	Line-ground resistance in Ω
$R{:}B = \Re(\underline{Z}_b)$	
$R{:}C = \Re(\underline{Z}_c)$	
$X{:}A = \Im(\underline{Z}_a)$	Line-ground network reactance in Ω
$X{:}B = \Im(\underline{Z}_b)$	
$X{:}C = \Im(\underline{Z}_c)$	
$Z{:}A = \lvert\underline{Z}_a\rvert$	Line-ground network impedance in Ω
$Z{:}B = \lvert\underline{Z}_b\rvert$	
$Z{:}C = \lvert\underline{Z}_c\rvert$	
$phiz{:}A = \phi_{\underline{Z}_a}$	Line-ground angle of network impedance in deg
$phiz{:}B = \phi_{\underline{Z}_b}$	
$phiz{:}C = \phi_{\underline{Z}_c}$	
$Rl{:}A = \Re(\underline{Z}l_a)$	Line-line resistance in Ω
$Rl{:}B = \Re(\underline{Z}l_b)$	
$Rl{:}C = \Re(\underline{Z}l_c)$	

$Xl{:}A = \Im(\underline{Zl}_a)$	Line-line network reactance in Ω
$Xl{:}B = \Im(\underline{Zl}_b)$	
$Xl{:}C = \Im(\underline{Zl}_c)$	
$Zl{:}A = \lvert\underline{Zl}_a\rvert$	Line-line network impedance in Ω
$Zl{:}B = \lvert\underline{Zl}_b\rvert$	
$Zl{:}C = \lvert\underline{Zl}_c\rvert$	
$phizl{:}A = \phi_{\underline{Z}l_a}$	Line-line angle of network impedance in deg
$phizl{:}B = \phi_{\underline{Z}l_b}$	
$phizl{:}C = \phi_{\underline{Z}l_c}$	
$Rln{:}A = \Re(\underline{Zln}_a)$	Line-neutral resistance in Ω
$Rln{:}B = \Re(\underline{Zln}_b)$	
$Rln{:}C = \Re(\underline{Zln}_c)$	
$Xln{:}A = \Im(\underline{Zln}_a)$	Line-neutral network reactance in Ω
$Xln{:}B = \Im(\underline{Zln}_b)$	
$Xln{:}C = \Im(\underline{Zln}_c)$	
$Zln{:}A = \lvert\underline{Zln}_a\rvert$	Line-neutral network impedance in Ω
$Zln{:}B = \lvert\underline{Zln}_b\rvert$	
$Zln{:}C = \lvert\underline{Zln}_c\rvert$	
$phizln{:}A = \phi_{\underline{Zln}_a}$	Line-neutral angle of network impedance in deg
$phizln{:}B = \phi_{\underline{Zln}_b}$	
$phizln{:}C = \phi_{\underline{Zln}_c}$	

$u0_{=}\sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{u0}_m\rvert^{\alpha}}$	Line-ground zero sequence voltage magnitude in p.u., where $\underline{u0}_m = \frac{1}{3}\left(\underline{u}_{a,m} + \underline{u}_{b,m} + \underline{u}_{c,m}\right)$
$U0 = u0 \cdot U_{nom}/\sqrt{3}$	Zero sequence voltage magnitude in kV
$U0x3 = 3 \cdot U0$	in kV
$phiu0$	(N/A) Zero sequence voltage angle in deg
$u1_{=}\sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{u1}_m\rvert^{\alpha}}$	Line-ground positive sequence voltage magnitude in p.u., according to IEC-61000-3-6 where $\underline{u1}_m = \frac{1}{3}(\underline{u}_{a,m} + a \cdot \underline{u}_{b,m} + a^2 \cdot \underline{u}_{c,m})$ and $a = \angle 120°$
$U1 = u1 \cdot U_{nom}/\sqrt{3}$	Line-ground positive sequence voltage magnitude in kV
$phiu1$	(N/A) Line-ground positive sequence voltage angle in deg
$u2_{=}\sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{u2}_m\rvert^{\alpha}}$	Line-ground negative sequence voltage magnitude in p.u., according to IEC-61000-3-6 where $\underline{u2}_m = \frac{1}{3}(\underline{u}_{a,m} + a^2 \cdot \underline{u}_{b,m} + a \cdot \underline{u}_{c,m})$ and $a = \angle 120°$
$U2 = u2 \cdot U_{nom}/\sqrt{3}$	Line-ground negative sequence voltage magnitude in kV
$phiu2$	(N/A) Line-ground negative sequence voltage angle v
$u1pc = u1 \cdot 100$	Line-ground positive sequence voltage magnitude in %
$U1l = u1 \cdot U_{nom}$	Line-line positive sequence voltage magnitude in kV
$U2l = u2 \cdot U_{nom}$	Line-line negative sequence voltage magnitude in kV
$un_{=}\sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{un}_m\rvert^{\alpha}}$	Neutral-ground voltage magnitude in p.u.
$Un = un \cdot U_{nom}/\sqrt{3}$	Neutral-ground voltage magnitude in kV
$R0 = \Re(\underline{Z0})$	Zero sequence resistance in Ω
$X0 = \Im(\underline{Z0})$	Zero sequence network reactance in Ω
$Z0 = \lvert\underline{Z0}\rvert$	Zero sequence network impedance in Ω
$phiz0 = \phi_{\underline{Z0}}$	Zero sequence network impedance angle in deg
$R1 = \Re(\underline{Z1})$	Positive sequence resistance in Ω
$X1 = \Im(\underline{Z1})$	Positive sequence network reactance in Ω
$Z1 = \lvert\underline{Z1}\rvert$	Positive sequence network impedance in Ω
$phiz1 = \phi_{\underline{Z1}}$	Positive sequence network impedance angle in deg

$R2 = \Re(\underline{Z2})$	Negative sequence resistance in Ω
$X2 = \Im(\underline{Z2})$	Negative sequence network reactance in Ω
$Z2 = \lvert\underline{Z2}\rvert$	Negative sequence network impedance in Ω
$phiz2 = \phi_{\underline{Z2}}$	Negative sequence network impedance angle in deg
$Rn = \Re(\underline{Z}_n)$	Resistance, neutral in Ω
$Xn = \Im(\underline{Z}_n)$	Network reactance, neutral in Ω
$Zn = \lvert\underline{Z}_n\rvert$	Network impedance, neutral in Ω
$phizn = \phi_{\underline{Z}_n}$	Angle of network impedance, neutral in Ω
$ubfac = \frac{u2}{u1}$	Unbalance factor

Monitor Variables Considering All Frequencies: The following monitor variables are calculated considering all frequencies, f_k:

Intermediate variables:

$u{:}A(f_k) = \sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{u}_{a,m}(f_k)\rvert^{\alpha}}$	Line-ground voltage, magnitude in p.u., according to IEC-61000-3-6
$u{:}B(f_k) = \sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{u}_{b,m}(f_k)\rvert^{\alpha}}$	
$u{:}C(f_k) = \sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{u}_{c,m}(f_k)\rvert^{\alpha}}$	
$u0(f_k) = \sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{u0}_m(f_k)\rvert^{\alpha}}$	Zero sequence voltage magnitude in p.u. where $\underline{u0}(f_k) = \frac{1}{3}\left(\underline{u}_{a,m}(f_k) + \underline{u}_{b,m}(f_k) + \underline{u}_{c,m}(f_k)\right)$
$u1(f_k) = \sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{u1}_m(f_k)\rvert^{\alpha}}$	Positive sequence voltage in p.u. where $\underline{u1}_m(f_k) = \frac{1}{3}(\underline{u}_{a,m}(f_k) + a\cdot\underline{u}_{b,m}(f_k) + a^2\cdot\underline{u}_{c,m}(f_k))$ and $a = \angle 120°$
$u2(f_k) = \sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{u2}_m(f_k)\rvert^{\alpha}}$	Negative sequence voltage in p.u. where $\underline{u2}_m(f_k) = \frac{1}{3}(\underline{u}_{a,m}(f_k) + a^2\cdot\underline{u}_{b,m}(f_k) + a\cdot\underline{u}_{c,m}(f_k))$ and $a = \angle 120°$
$ul{:}A(f_k) = \sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{ul}_{a,m}(f_k)\rvert^{\alpha}}$	Line-line voltage magnitude in p.u.
$ul{:}B(f_k) = \sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{ul}_{b,m}(f_k)\rvert^{\alpha}}$	
$ul{:}C(f_k) = \sqrt{[\alpha]\sum_{m=0}^{N}\lvert\underline{ul}_{c,m}(f_k)\rvert^{\alpha}}$	

Monitor variables:

$urms{:}A = \sqrt{\sum_{k\geq 1} u{:}A(f_k)^2}$	RMS value of line-neutral voltage in p.u.
$urms{:}B = \sqrt{\sum_{k\geq 1} u{:}B(f_k)^2}$	
$urms{:}C = \sqrt{\sum_{k\geq 1} u{:}C(f_k)^2}$	
$Urms{:}A = urms{:}A \cdot U_{nom}/\sqrt{3}$	RMS value of line-neutral voltage in kV
$Urms{:}B = urms{:}B \cdot U_{nom}/\sqrt{3}$	
$Urms{:}C = urms{:}C \cdot U_{nom}/\sqrt{3}$	
$u0rms = \sqrt{\sum_{k\geq 1} u0(f_k)^2}$	RMS value of zero sequence voltage in p.u.
$U0rms = u0rms \cdot U_{nom}/\sqrt{3}$	RMS value of zero sequence voltage in kV
$u1rms = \sqrt{\sum_{k\geq 1} u1(f_k)^2}$	RMS value of positive sequence voltage in p.u.
$U1rms = u1rms \cdot U_{nom}/\sqrt{3}$	RMS value of positive sequence voltage in kV
$U1lrms = u1rms \cdot U_{nom}$	RMS value of positive sequence line-line voltage in kV
$u2rms = \sqrt{\sum_{k\geq 1} u2(f_k)^2}$	RMS value of negative sequence voltage in p.u.
$U2rms = u2rms \cdot U_{nom}/\sqrt{3}$	RMS value of negative sequence voltage in kV
$U2lrms = u2rms \cdot U_{nom}$	RMS value of negative sequence line-line voltage in kV
$HD{:}A = \frac{u{:}A(f_{out})}{u{:}A(f_1)} \cdot 100$	Harmonic distortion in %; based on fundamental frequency values
$HD{:}B = \frac{u{:}B(f_{out})}{u{:}B(f_1)} \cdot 100$	
$HD{:}C = \frac{u{:}C(f_{out})}{u{:}C(f_1)} \cdot 100$	

$HD{:}A = \frac{u{:}A(f_{out})}{u_{nom}{:}A} \cdot 100$	Harmonic distortion in %; based on nominal voltage, where $u_{nom} = 1$p.u.
$HD{:}B = \frac{u{:}B(f_{out})}{u_{nom}{:}B} \cdot 100$	
$HD{:}C = \frac{u{:}C(f_{out})}{u_{nom}{:}C} \cdot 100$	
$HF{:}A = \frac{u{:}A(f_{out})}{urms{:}A} \cdot 100$	Harmonic factor in %
$HF{:}B = \frac{u{:}B(f_{out})}{urms{:}B} \cdot 100$	
$HF{:}C = \frac{u{:}C(f_{out})}{urms{:}C} \cdot 100$	
$HD0 = \frac{u0(f_{out})}{\sqrt{u0(f_1)^2+u1(f_1)^2+u2(f_1)^2}} \cdot 100$	Harmonic distortion, zero sequence in %; based on fundamental frequency values
$HD1 = \frac{u1(f_{out})}{\sqrt{u0(f_1)^2+u1(f_1)^2+u2(f_1)^2}} \cdot 100$	Harmonic distortion, positive sequence in %
$HD2 = \frac{u2(f_{out})}{\sqrt{u0(f_1)^2+u1(f_1)^2+u2(f_1)^2}} \cdot 100$	Harmonic distortion, negative sequence in %
$HD0 = \frac{u0(f_{out})}{u_{nom}} \cdot 100$	Harmonic distortion, zero sequence in %; based on nominal voltage
$HD1 = \frac{u1(f_{out})}{u_{nom}} \cdot 100$	Harmonic distortion, positive sequence in %
$HD2 = \frac{u2(f_{out})}{u_{nom}} \cdot 100$	Harmonic distortion, negative sequence in %
$HF0 = \frac{u0(f_{out})}{u0rms} \cdot 100$	Harmonic factor, zero sequence in %
$HF1 = \frac{u1(f_{out})}{u1rms} \cdot 100$	Harmonic factor, positive sequence in %
$HF2 = \frac{u2(f_{out})}{u2rms} \cdot 100$	Harmonic factor, negative sequence in %

$THD{:}A = \frac{1}{u{:}A(f_1)} \sqrt{urms{:}A^2 - u{:}A(f_1)^2} \cdot 100$	Total harmonic distortion in %; based on fundamental frequency values
$THD{:}B = \frac{1}{u{:}B(f_1)} \sqrt{urms{:}B^2 - u{:}B(f_1)^2} \cdot 100$	
$THD{:}C = \frac{1}{u{:}C(f_1)} \sqrt{urms{:}C^2 - u{:}C(f_1)^2} \cdot 100$	
$THD{:}A = \frac{1}{u_{nom}{:}A} \sqrt{urms{:}A^2 - u{:}A(f_1)^2} \cdot 100$	Total harmonic distortion in %; based on nominal voltage, where $u_{nom} = 1$p.u.
$THD{:}B = \frac{1}{u_{nom}{:}B} \sqrt{urms{:}B^2 - u{:}B(f_1)^2} \cdot 100$	
$THD{:}C = \frac{1}{u_{nom}{:}C} \sqrt{urms{:}C^2 - u{:}C(f_1)^2} \cdot 100$	
$THF{:}A = \frac{1}{urms{:}A} \sqrt{urms{:}A^2 - u{:}A(f_1)^2} \cdot 100$	Total harmonic factor in %
$THF{:}B = \frac{1}{urms{:}B} \sqrt{urms{:}B^2 - u{:}B(f_1)^2} \cdot 100$	
$THF{:}C = \frac{1}{urms{:}C} \sqrt{urms{:}C^2 - u{:}C(f_1)^2} \cdot 100$	
$ulrms{:}A = \sqrt{\sum_{k \geq 1} ul{:}A(f_k)^2}$	RMS value of line-line voltage in p.u.
$ulrms{:}B = \sqrt{\sum_{k \geq 1} ul{:}B(f_k)^2}$	
$ulrms{:}C = \sqrt{\sum_{k \geq 1} ul{:}C(f_k)^2}$	
$Ulrms{:}A = ulrms{:}A \cdot U_{nom}$	RMS value of line-line voltage in kV
$Ulrms{:}B = ulrms{:}B \cdot U_{nom}$	
$Ulrms{:}C = ulrms{:}C \cdot U_{nom}$	
$TAD{:}A = \frac{1}{u{:}A(f_1)} \cdot \left[\sum_{k \geq 1} u{:}A(f_k) - u{:}A(f_1)\right] \cdot 100$	Total arithmetic distortion in %
$TAD{:}B = \frac{1}{u{:}B(f_1)} \cdot \left[\sum_{k \geq 1} u{:}B(f_k) - u{:}B(f_1)\right] \cdot 100$	
$TAD{:}C = \frac{1}{u{:}C(f_1)} \cdot \left[\sum_{k \geq 1} u{:}C(f_k) - u{:}C(f_1)\right] \cdot 100$	
$uasum{:}A = \sum_{k \geq 1} u{:}A(f_k)$	Arithmetic voltage sum in p.u.
$uasum{:}B = \sum_{k \geq 1} u{:}B(f_k)$	
$uasum{:}C = \sum_{k \geq 1} u{:}C(f_k)$	

$TIF{:}A = \frac{1}{urms{:}A}\sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot u{:}A(f_k)^2}$ Telephone interference factor

$TIF{:}B = \frac{1}{urms{:}B}\sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot u{:}B(f_k)^2}$

$TIF{:}C = \frac{1}{urms{:}C}\sqrt{\sum_{k\geq 1} tif(f_k)^2 \cdot u{:}C(f_k)^2}$

$urmsint = \max_{x=[A,B,C]}(urmsint{:}x)$ RMS value of integer harmonics in p.u., where $urmsint{:}x = \sqrt{\sum_{\substack{k\geq 1 \\ integer}} u{:}x(f_k)^2}$

$urmsnint = \max_{x=[A,B,C]}(urmsnint{:}x)$ RMS value of non-integer harmonics in p.u., where $urmsnint{:}x = \sqrt{\sum_{\substack{k\geq 1 \\ non-integer}} u{:}x(f_k)^2}$

$THD0 = \frac{1}{u1(f_1)^2+u0(f_1)^2+u2(f_1)^2}\sqrt{u0rms^2 - u0(f_1)^2}$ Total harmonic distortion in % (voltage, zero sequence)

$THD1 = \frac{1}{u1(f_1)^2+u0(f_1)^2+u2(f_1)^2}\sqrt{u1rms^2 - u1(f_1)^2}$ Total harmonic distortion in % (voltage, positive sequence)

$THD2 = \frac{1}{u1(f_1)^2+u0(f_1)^2+u2(f_1)^2}\sqrt{u2rms^2 - u2(f_1)^2}$ Total harmonic distortion in % (voltage, negative sequence)

$THDbal = \sqrt{THD1^2 + THD2^2}$ Total harmonic distortion in %, excluding zero sequence

$THDtot = \sqrt{THD1^2 + THD2^2 + THD0^2}$ Total harmonic distortion in %, including zero sequence

$TADmx = \max_{x=[A,B,C]}(TAD{:}x)$ Arithmetic distortion in %

$TADint = \max_{x=[A,B,C]}(TADint{:}x)$ Arithmetic distortion in %, where

$$TADint{:}x = \frac{1}{u{:}x(f_1)} \cdot \left[\sum_{\substack{k\geq 1 \\ integer}} u{:}x(f_k) - u{:}x(f_1)\right] \cdot 100$$

$TADnint = \max_{x=[A,B,C]}(TADnint{:}x)$ Arithmetic distortion in %, where

$$TADnint{:}x = \frac{1}{u{:}x(f_1)} \cdot \left[\sum_{\substack{k\geq 1 \\ non-integer}} u{:}x(f_k) - u{:}x(f_1)\right] \cdot 100$$

$TAD_3 = \max_{x=[A,B,C]}(TADint{:}x)$ Arithmetic distortion in %, harmonic orders: multiples of three

$TAD_2 = \max_{x=[A,B,C]}(TADint{:}x)$ Arithmetic distortion in %, harmonic orders: multiples of two

$THDint = \max_{x=[A,B,C]}(THDint{:}x)$

Total harmonic distortion in %, where

$THDint{:}x = \frac{1}{u{:}x(f_1)} \cdot \sqrt{urms{:}x^2 - u{:}x(f_1)^2} \cdot 100$

(Fundamental frequency values) or

$THDint{:}x = \frac{1}{u_{nom}{:}x} \cdot \sqrt{urms{:}x^2 - u{:}x(f_1)^2} \cdot 100$

(Nominal voltage) and

$urms{:}x = \sqrt{\sum_{\substack{k\geq 1 \\ integer}} u{:}x(f_k)^2}$

$THDnint = \max_{x=[A,B,C]}(THDnint{:}x)$

Total harmonic distortion in %, where

$THDnint{:}x = \frac{1}{u{:}x(f_1)} \cdot \sqrt{urms{:}x^2 - u{:}x(f_1)^2} \cdot 100$

(Fundamental frequency values) or

$THDnint{:}x = \frac{1}{u_{nom}{:}x} \cdot \sqrt{urms{:}x^2 - u{:}x(f_1)^2} \cdot 100$

(Nominal voltage) and

$urms{:}x = \sqrt{\sum_{\substack{k>1 \\ non-integer}} u{:}x(f_k)^2}$

$THD_3 = \max_{x=[A,B,C]}(THDint{:}x)$

Total harmonic distortion in %, harmonic orders: multiples of three

$THD_2 = \max_{x=[A,B,C]}(TADint{:}x)$

Total harmonic distortion in %, harmonic orders: multiples of two

$TIFmx = \max_{x=[A,B,C]}(TIF{:}x)$

Total interference factor, for harmonic orders

$uasummx = \max_{x=[A,B,C]}(uasum{:}x)$

Arithmetic voltage sum in p.u.

$uasumint = \max_{x=[A,B,C]}(uasumint{:}x)$

Arithmetic sum of integer harmonics in p.u., where

$uasumint{:}x = \sum_{\substack{k\geq 2 \\ integer}} u{:}x(f_k)$

$uasumnint = \max_{x=[A,B,C]}(uasumnint{:}x)$

Arithmetic sum of integer harmonics in p.u., where

$uasumnint{:}x = \sum_{\substack{k>2 \\ non-integer}} u{:}x(f_k)$

4 Calculation Considering IEC Flicker Sources

PowerFactory calculates monitor variables for the Flicker Assessment calculation within the Harmonic Load Flow command if at least one flicker source is identified in the network. Such sources may include asynchronous machines (*ElmAsm*), doubly-fed asynchronous machines (*ElmAsmsc*) and static generators (*ElmGenstat*).

The calculation of flicker sources in the network is always based on the fundamental frequency. Quantities are computed for busbars only. Table 4.1 provides the nomenclature used in this section; detailed derivation of these quantities can be found in [1].

Table 4.1: Nomenclature

Name	Unit	Description
ψ_k	deg	Network impedance angle, as seen by WTG k
v_a	m/s	Annual average wind speed in m/s
$c(\psi_k,v_a)$		Flicker coefficient for continuous operation of the WTG
S_n	VA	Rated apparent power of WTG
S_k	VA	Short-circuit apparent power of grid
N_{wt}	-	Number of WTGs connected to PCC
i	-	The ith WTG
N_{10}	-	Maximum number of switching operations within a 10-minute period
N_{120}	-	Maximum number of switching operations within a 120-minute period
k_f	-	Flicker step factor
k_u	-	Voltage change factor

4.1 Balanced and Unbalanced Calculations

4.1.0.1 Bus Monitor Variables

$P_{st\Sigma_c} = \frac{1}{S_k} \cdot \sqrt{\sum_{i=1}^{N_{wt}} \left(c_i \left(\psi_k, v_a\right) \cdot S_{n,i}\right)^2}$	Short-term flicker disturbance factor (continuous operation of the WTG)
$P_{lt\Sigma_c} = \frac{1}{S_k} \cdot \sqrt{\sum_{i=1}^{N_{wt}} \left(c_i \left(\psi_k, v_a\right) \cdot S_{n,i}\right)^2}$	Long-term flicker disturbance factor (continuous operation of the WTG)
$P_{st\Sigma_s} = \frac{18}{S_k} \cdot \left[\sum_{i=1}^{N_{wt}} N_{10,i} \cdot \left(k_{f,i}(\psi_k) \cdot S_{n,i}\right)^{3.2}\right]^{0.31}$	Short-term flicker disturbance factor (switching operation of the WTG)
$P_{lt\Sigma_s} = \frac{8}{S_k} \cdot \left[\sum_{i=1}^{N_{wt}} N_{120,i} \cdot \left(k_{f,i}(\psi_k) \cdot S_{n,i}\right)^{3.2}\right]^{0.31}$	Short-term flicker disturbance factor (switching operation of the WTG)
$d = 100 \cdot k_u(\psi_k) \cdot \frac{S_n}{S_k}$	Relative voltage change due to the switching operation of a single WTG in % (Note: only maximum for entire network reported)

5 Frequency Dependent Mutual Data

The *Frequency Dependent Mutual Data* element is used to represent mutual impedances between buses.

5.1 Definition

A *Frequency Dependent Mutual Data* element can be defined by selecting two buses, right-clicking and selecting *Define... Freq. Dep. Mutual Data* from the context-sensitive menu. The element dialog will then be displayed, with both buses assigned. No additional inputs are required. These elements are used to represent mutual impedances between buses. Impedance-related calculation variables are available for bus 1-2 and bus 2-1.

It is noted that the presence of phase-shifting transformers in the system can lead to different coupling impedances for bus 1-2 and bus 2-1.

5.2 Harmonic Load Flow and Frequency Sweep

5.2.1 Balanced

Table 5.1: Calculation variables

Name	Unit	Description
bus1		Node 1
bus2		Node 2
R_12	Ω	Coupling Resistance Node 1 - Node 2
X_12	Ω	Coupling Reactance Node 1 - Node 2
Z_12	Ω	Coupling Impedance Node 1 - Node 2
phiz_12	deg	Coupling Impedance Node 1 - Node 2, Angle
R_21	Ω	Coupling Resistance Node 2 - Node 1
X_21	Ω	Coupling Reactance Node 2 - Node 1
Z_21	Ω	Coupling Impedance Node 2 - Node 1
phiz_21	deg	Coupling Impedance Node 2 - Node 1, Angle

5.2.2 Unbalanced

Table 5.2: Calculation variables

Name	Unit	Description
bus1		Node 1
bus2		Node 2
R_12	Ω	Coupling Resistance Node 1 - Node 2
X_12	Ω	Coupling Reactance Node 1 - Node 2
Z_12	Ω	Coupling Impedance Node 1 - Node 2
phiz_12	Ω	Coupling Impedance Node 1 - Node 2, Angle
R_21	Ω	Coupling Resistance Node 2 - Node 1
X_21	Ω	Coupling Reactance Node 2 - Node 1
Z_21	Ω	Coupling Impedance Node 2 - Node 1
phiz_21	Ω	Coupling Impedance Node 2 - Node 1, Angle
R0_12	Ω	Coupling Resistance Node 1 - Node 2, 0-Sequ.
X0_12	Ω	Coupling Reactance Node 1 - Node 2, 0-Sequ.
Z0_12	Ω	Coupling Impedance Node 1 - Node 2, 0-Sequ.
phiz0_12	Ω	Coupling Impedance Node 1 - Node 2, Angle., 0-Sequ.
R0_21	Ω	Coupling Resistance Node 2 - Node 1, 0-Sequ.
X0_21	Ω	Coupling Reactance Node 2 - Node 1, 0-Sequ.
Z0_21	Ω	Coupling Impedance Node 2 - Node 1, 0-Sequ.
phiz0_21	Ω	Coupling Impedance Node 2 - Node 1, Angle., 0-Sequ.
R1_12	Ω	Coupling Resistance Node 1 - Node 2, 1-Sequ.
X1_12	Ω	Coupling Reactance Node 1 - Node 2, 1-Sequ.
Z1_12	Ω	Coupling Impedance Node 1 - Node 2, 1-Sequ.
phiz1_12	Ω	Coupling Impedance Node 1 - Node 2, Angle., 1-Sequ.
R1_21	Ω	Coupling Resistance Node 2 - Node 1, 1-Sequ.
X1_21	Ω	Coupling Reactance Node 2 - Node 1, 1-Sequ.
Z1_21	Ω	Coupling Impedance Node 2 - Node 1, 1-Sequ.
phiz1_21	Ω	Coupling Impedance Node 2 - Node 1, Angle., 1-Sequ.
R2_12	Ω	Coupling Resistance Node 1 - Node 2, 2-Sequ.
X2_12	Ω	Coupling Reactance Node 1 - Node 2, 2-Sequ.
Z2_12	Ω	Coupling Impedance Node 1 - Node 2, 2-Sequ.
phiz2_12	Ω	Coupling Impedance Node 1 - Node 2, Angle., 2-Sequ.
R2_21	Ω	Coupling Resistance Node 2 - Node 1, 2-Sequ.
X2_21	Ω	Coupling Reactance Node 2 - Node 1, 2-Sequ.
Z2_21	Ω	Coupling Impedance Node 2 - Node 1, 2-Sequ.
phiz2_21	Ω	Coupling Impedance Node 2 - Node 1, Angle., 2-Sequ.

6 Frequency Dependent Network Equivalent Data

The *Frequency Dependent Network Equivalent Data* element is used to represent equivalent impedances and admittances between buses.

6.1 Definition

A *Frequency Dependent Network Equivalent Data* element (ElmEquivalent) can be defined by right-clicking on multiple buses and selecting *Define... Freq. Dep. Network Equivalent Data* from the context-sensitive menu. The element dialog will then be displayed, with the bus/es as an entry in the table. No additional inputs are required. The purpose of this element is to provide a means to access the frequency dependent network equivalent self- and mutual impedances and admittances of the selected buses. It is available for use in the Frequency Sweep and the calculation results are the impedance and admittance matrices that represent the equivalent system of user-selected buses. In addition, a radiality factor (RF) can be calculated, as defined in [2].

6.2 Frequency Sweep

Frequency Dependent Network Equivalent Data elements are used to represent self-impedances and admittances of buses, and mutual impedances and admittances between buses. Impedance- and admittance related calculation variables are available in matrix form, representing the equivalent defined by the selected buses. These matrices must be accessed using a zero-based index, and the buses are ordered in the matrices in the order that they appear in the equivalent element table. Assuming that two buses have been selected, an example configuration of results is:

Z:0:0:r	The real part of the self-impedance in Ω at the first bus (index '0') defined in the equivalent element.
Z:0:1:mag	The magnitude of the mutual impedance in Ω between the first (index '0') and second buses (index '1') defined in the equivalent element.
Y:1:1:phi	The angle of the self-admittance in deg of the second bus (index '1') defined in the equivalent element.

It is noted that the presence of phase-shifting transformers in the system can lead to different coupling impedances for bus 1-2 and bus 2-1.

The selection of a single "Oscillating generator" and a single "Influencing device" is only required for the calculation of the radiality factor.

6.2.1 Balanced

Table 6.1: Calculation variables

Name	Unit	Description
buses		Nodes
Z:r	Ω	Equivalent impedance matrix, Real Part
Z:i	Ω	Equivalent impedance matrix, Imaginary Part
Z:phi	deg	Equivalent impedance matrix, Angle
Z:mag	Ω	Equivalent impedance matrix, Magnitude
Z:phirad	rad	Equivalent impedance matrix, Angle
Y:r	S	Equivalent admittance matrix, Real Part
Y:i	S	Equivalent admittance matrix, Imaginary Part
Y:phi	deg	Equivalent admittance matrix, Angle
Y:mag	S	Equivalent admittance matrix, Magnitude
Y:phirad	rad	Equivalent admittance matrix, Angle
RF		Radiality factor

6.2.2 Unbalanced

Table 6.2: Calculation variables

Name	Unit	Description
buses		Nodes
Z:r	Ω	Equivalent impedance matrix, Real Part
Z:i	Ω	Equivalent impedance matrix, Imaginary Part
Z:phi	deg	Equivalent impedance matrix, Angle
Z:mag	Ω	Equivalent impedance matrix, Magnitude
Z:phirad	rad	Equivalent impedance matrix, Angle
Y:r	S	Equivalent admittance matrix, Real Part
Y:i	S	Equivalent admittance matrix, Imaginary Part
Y:phi	deg	Equivalent admittance matrix, Angle
Y:mag	S	Equivalent admittance matrix, Magnitude
Y:phirad	rad	Equivalent admittance matrix, Angle
RF		Radiality factor

7 References

[1] IEC 61400 Part 21: Measurement and Assessment of Power Quality Characteristics of Grid Connected Wind Turbines, 2001.

[2] M. S. Annakkage, C. Karawita, and U. D. Annakkage. Frequency scan-based screening method for device dependent sub-synchronous oscillations. *IEEE Transactions on Power Systems*, 31(3):1872–1878, 2016.

[3] Standards Australia. AS/NZS61000.3.6:2001 Electromagnetic compatibility (EMC) - Limits - Assessment of emission limits for distorting loads in MV and HV power systems (IEC 61000-3-6:1996).